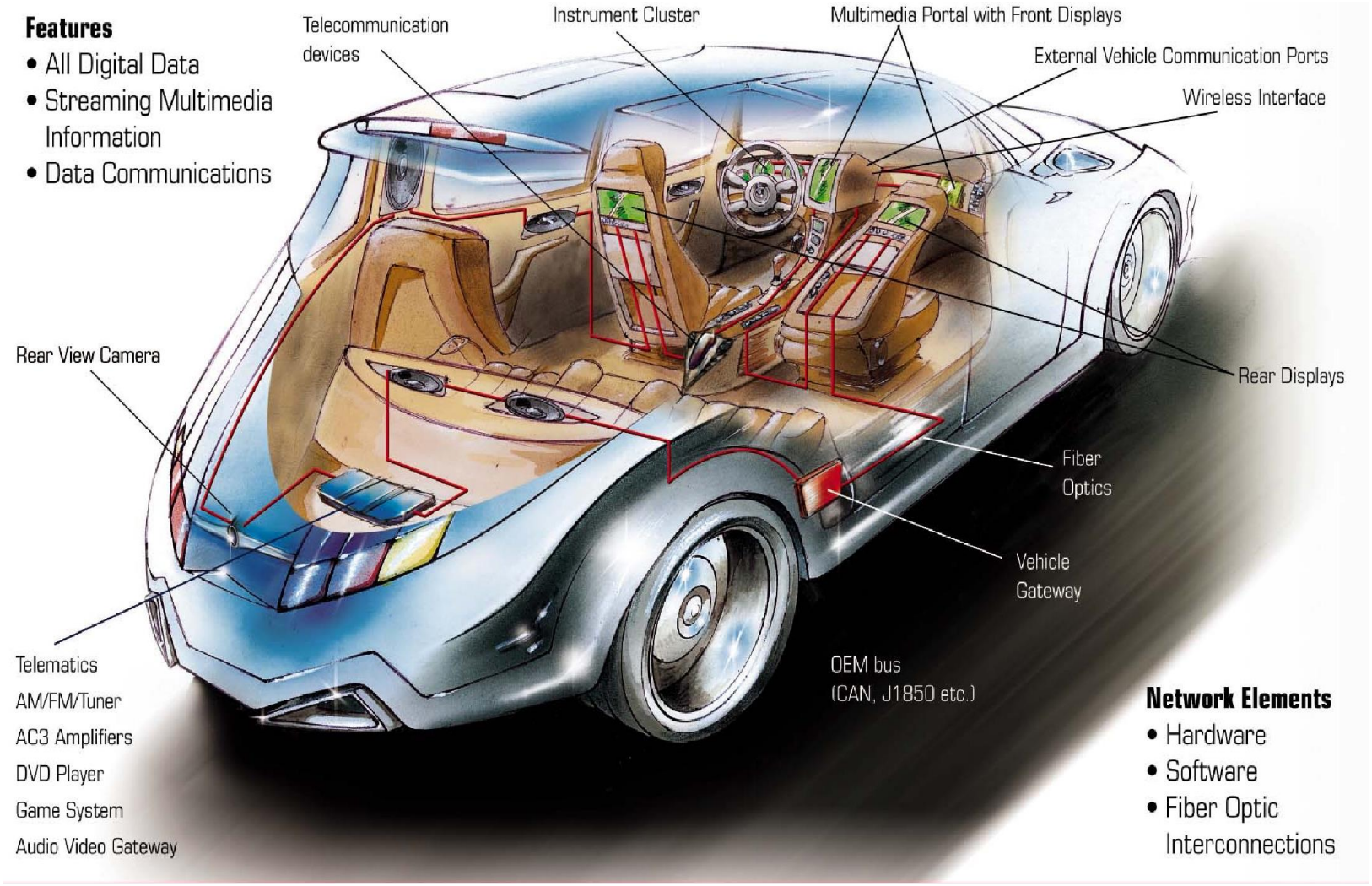
§5.3 网络通讯/ Network and Communication





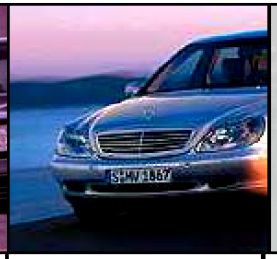
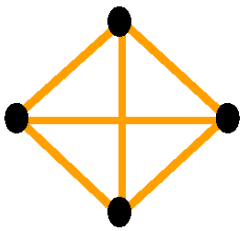
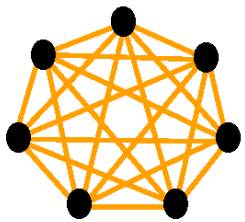
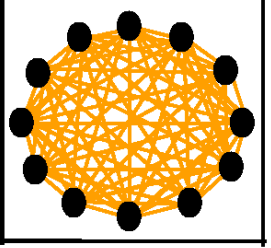
§5.3 网络通讯/ Network and Communication

- Introduction/简介
- Classifications of Bus System in Vehicle
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- Applications of TTCAN



§5.3 网络通讯/ Network and Communication

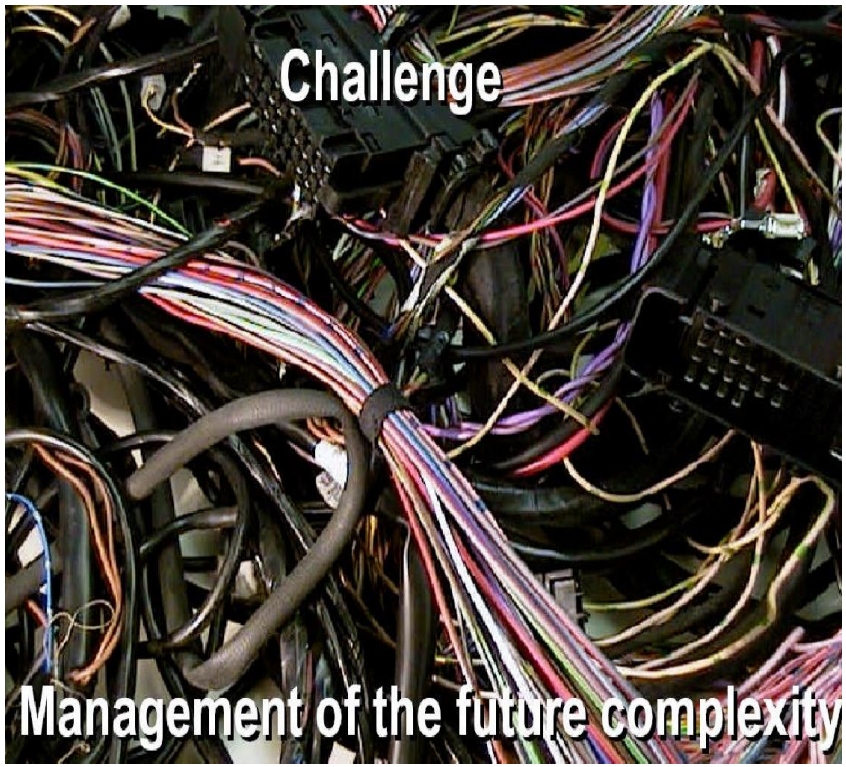
- Introduction(1): Why do we need Bus System?
  - More and More Electronic and Electric Systems are Added into the Car.

			
1979	1991	1998	2002
Number of Multimedia Components			
2	4	7	12
Number of Possible Interconnections			
1 	6 	28 	66 



§5.3 网络通讯/ Network and Communication

- Disadvantage of conventional wiring
  - Weight, Risk of failure , Electromagnetic incompatibility, Costs
- Solutions: Bus Systems
  - Serial communication via two–wire data line;
  - Bus control logic for:
    - Packaging and sending of data,
    - Bus guarding,
    - Verification of the correctness of data–transmission.
- Advantages:
  - Low Weight, More Safety,
  - Add–on, Function Relocation,……





§5.3 网络通讯/ Network and Communication

BODY-CAN

1. Zentralgateway (ZGW)
2. Elektronisches Zündschloss (EZS)
3. Türsteuergerät vorne links (TSG VL)
4. Türsteuergerät vorne rechts (TSG VR)
5. Türsteuergerät hinten links (TSG HL)
6. Türsteuergerät hinten rechts (TSG HR)
7. Sitzverstellung Fahrer (SSG-F)
8. Sitzverstellung Beifahrer (SSG-BF)
9. Sitzverstellung Fond (SSG-Fond)
10. Multikonturlehne vorne Links (MKL-VL)
11. Multikonturlehne vorne rechts (MKL-VR)
12. Multikonturlehne hinten Links (MKL-HL)
13. Multikonturlehne hinten rechts (MKL-HR)
14. Dachbedieneinheit (DBE)
15. SAM/SRB vorne
16. SAM/SRB hinten
17. Oberes-Bedien-Feld (OBF)
18. Reifendruckkontrolle (RDK)
19. Pumpe Multikontur (FDS)
20. Anhänger Anschlussgerät (AAG)
21. Parktronicsystem (PTS)
22. Multifunktionssteuergerät (MSS)
23. Standheizung (STH)
24. Heckdeckelfernschließung (HFS)
25. Keyless Go (KG)
26. Klimatisierungsautomatik (KLA)
27. Rückfahrkamera

CHASSIS-CAN

1. Zentralgateway (ZGW)
2. Elektronisches Zündschloss (EZS)
28. Motorelektronik (Benziner)
29. Motorelektronik (Diesel)
30. Sensor-Brake-Control (SBC)
31. Radar Decision Unit (RDU)
32. Mantelrohrmodul (MRSM)
33. Bordnetzsteuergerät (BNS)
34. Airbag ARCADE
35. Reversibler Gurtstraffer vorne links
36. Reversibler Gurtstraffer vorne rechts
37. Active-Body-Control (ABC)

DYNAMICS-CAN

30. Sensor-Brake-Control (SBC)
31. Radar Decision Unit (RDU)
38. Sensorcluster
39. Distronic (DTR)
40. Nightview

POWERTRAIN-CAN

28. Motorelektronik (Benziner)
29. Motorelektronik (Diesel)
41. Elektronische-Getr.-Steuerung (NAG2)
42. Intelligentes Servomodul (ISM)
43. Tanksteuergerät (nur M275)
44. Elektronische-Getr.-Steuerung (EGS 52 nur M275)

MOST-RING

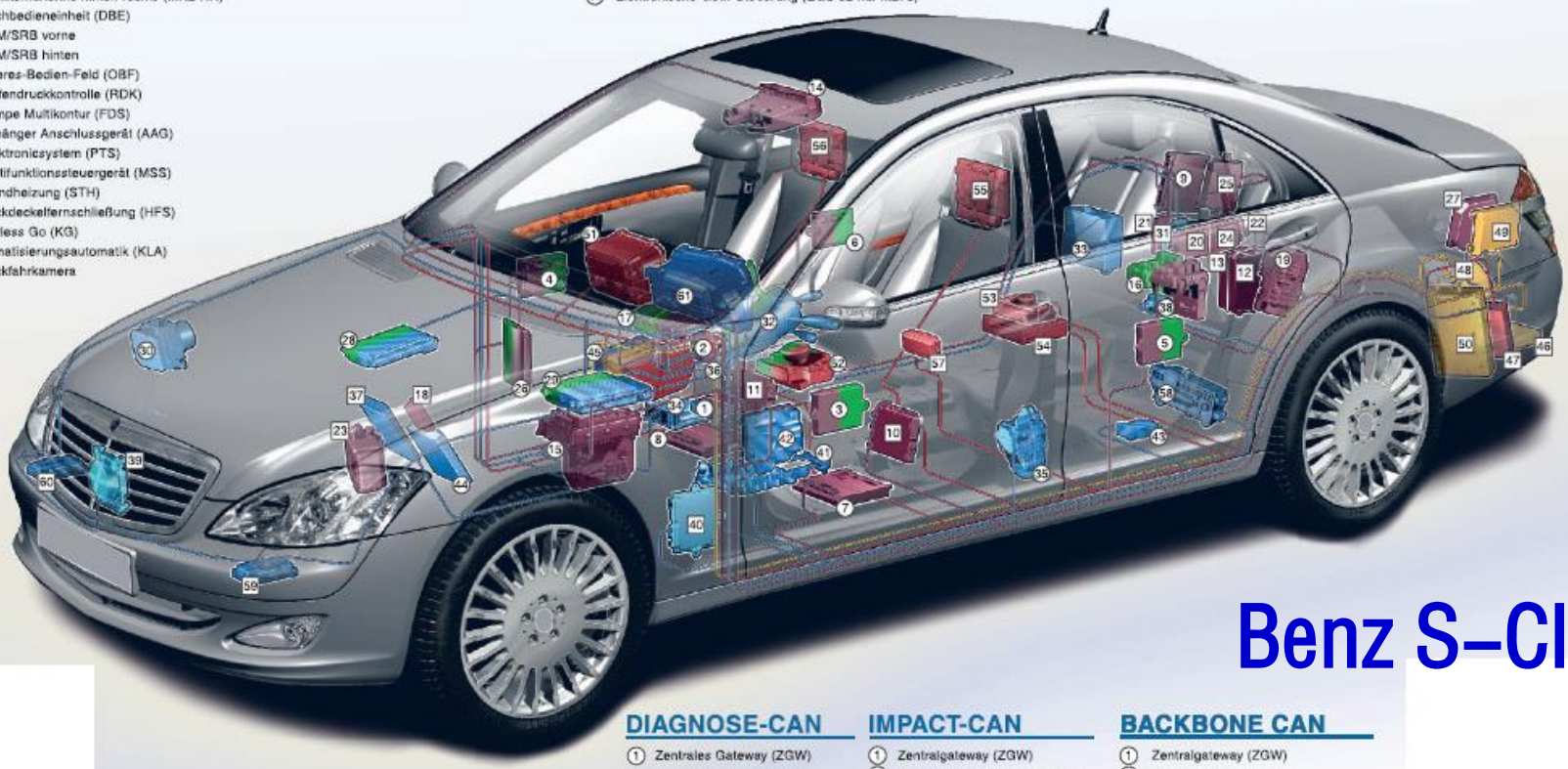
45. COMAND (TGW)
46. SDARS (nur USA)
47. TV-Tuner
48. Sprach-Bedien-System (SDS)
49. Universal Handy Interface (UHI)
50. Soundverstärker

HEADUNIT-CAN

45. COMAND (TGW)
47. TV-Tuner
51. Central Display
52. Dreh-Drücksteller
53. Dreh-Drücksteller hinten
54. Audio-Video-Splitter (AVS)
55. Fond-Display links
56. Fond-Display rechts
57. Fond Klima

LIN-BUS

3. Türsteuergerät vorne links (TSG VL)
4. Türsteuergerät vorne rechts (TSG VR)
5. Türsteuergerät hinten links (TSG HL)
6. Türsteuergerät hinten rechts (TSG HR)
16. SAM/SRB hinten
17. Oberes-Bedien-Feld (OBF)
26. Klimatisierungsautomatik (KLA)
28. Motorelektronik (Benziner)
29. Motorelektronik (Diesel)
32. Mantelrohrmodul (MRSM)
52. Dreh-Drücksteller



DIAGNOSE-CAN

1. Zentrales Gateway (ZGW)

IMPACT-CAN

1. Zentralgateway (ZGW)
58. Elektronische Feststellbremse
59. Headlamp Module Left (xalva)
60. Headlamp Module right (xalva)

BACKBONE CAN

1. Zentralgateway (ZGW)
45. COMAND (TGW)
61. Kombiinstrument (KI)

Σ aller Steuergeräte: 61

○ Serienausstattung
□ Sonderausstattung

Benz S–Class



§5.3 网络通讯/ Network and Communication

- 70 Computing Devices
- 2 Optical networks (MOST and Byteflight)
- 3 CAN networks (low–and high–speed)
- 3 special purpose sub–networks
- More than 5000 different Messages





§5.3 网络通讯/ Network and Communication

- SAE J1850 (1) VPWM, PWM
- LIN (Local Interconnect Network)
- CAN (Controller Area Network)
- MVB (Multipurpose Vehicle Bus)
- LON (Local Operating Network)
- TTP/C (Time Triggered Protocol)/ Class C
- Byteflight (Optimal Fiber Network)
- Flexray (Under Development)
- MOST (Media Oriented Systems Transport)
- TTCAN (Time Triggered CAN)
- Bluetooth (Wireless Network)









§5.3 网络通讯/ Network and Communication

Demands of Bus Systems

- High Speed: → CAN, MVB, TTP/C, Flexray;
  - Control devices e.g. for motors, gear, bogie and brakes;
  - Data rates between 200kbit/s and 25Mbit/s (For real time control);
- Electronics for car body and convenience;
  - Central locking, air–condition, adjustment of seats and mirrors, automatization of freight traffic;
  - Low price of components and wiring are more important than high transmission rates;
- Mobile Communication :
  - Audio, Video, telecommunication;
  - High transmission rates up to 24.8Mbit/s, low costs.
- Diagnosis



§5.3 网络通讯/ Network and Communication

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§5.3 网络通讯/ Network and Communication

●SAE Defined 3 Classes for Different Application:

- ***Class C Communications:*** 100kbit/s< C <1Mbit/s
  - Controller Area Network - High Speed (**CAN**)
- ***Class B Communications:*** 10kbit/s< B <100kbit/s
  - ***SAE J1850***
    - GM Class 2
    - Chrysler PCI-MS
    - Ford SCP
  - Controller Area Network - 'Low' Speed, Fault Tolerant (**CAN**)
- ***Class A Communications:*** <10kbit/s
  - Local Interconnect Network (**LIN**)
  - Distributed Systems Interface (**DSI**)
  - IIC, UBP, Sinebus, J1708 / J1587 / J1922, BEAN, and….and….and….

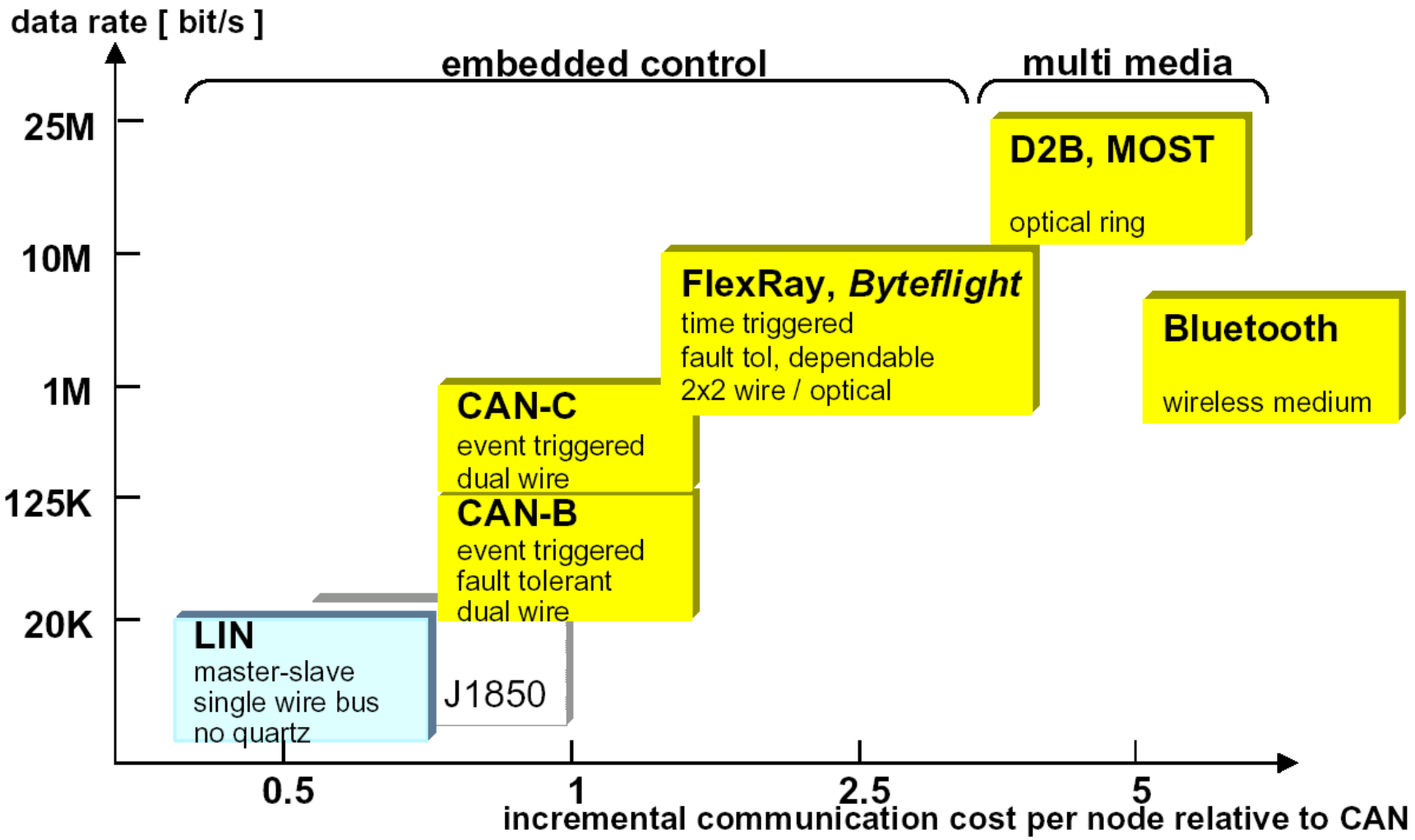


§5.3 网络通讯/ Network and Communication

- Class A networks are used for non–critical applications such as control of interior lights diagnostics etc. The volume of transferred data and the required transmission speed is low;
- Class B networks are used for applications that are important but not essential for proper vehicle operation, e.g. display of vehicle status information as current speed, fuel level etc. The volume of transferred data is moderate and the required transmission speed is medium;
- Class C networks are used for safety critical distributed real time application such as steering control, engine control and antilock brakes etc. The volume of transferred data is high and short latency times are required thus the transmission speed must be high.



§5.3 网络通讯/ Network and Communication



CAN and LIN are the Most Popular Bus System
FlexRay and MOST Have Been Put Into Mass Production



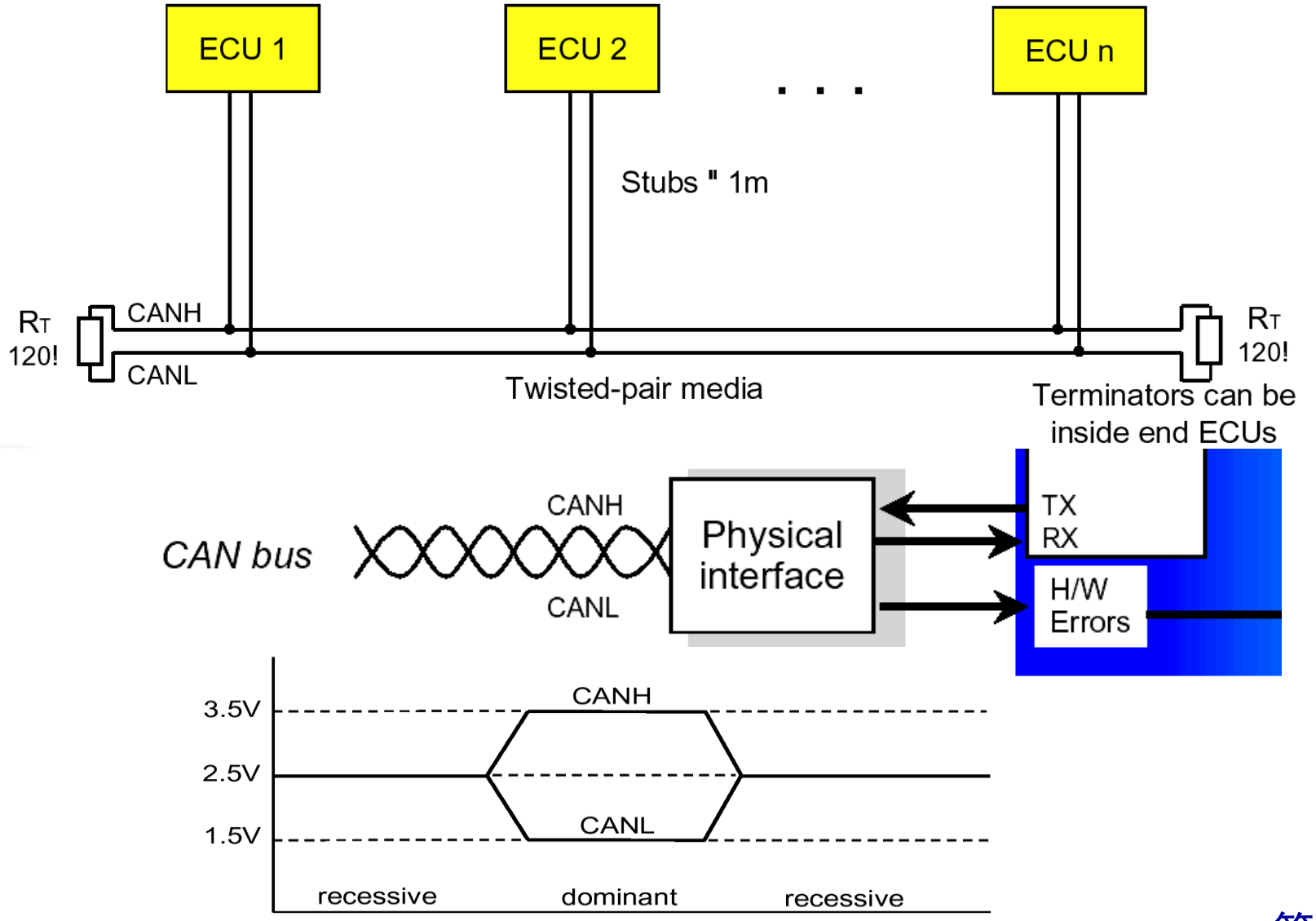
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§5.3 网络通讯/ Network and Communication

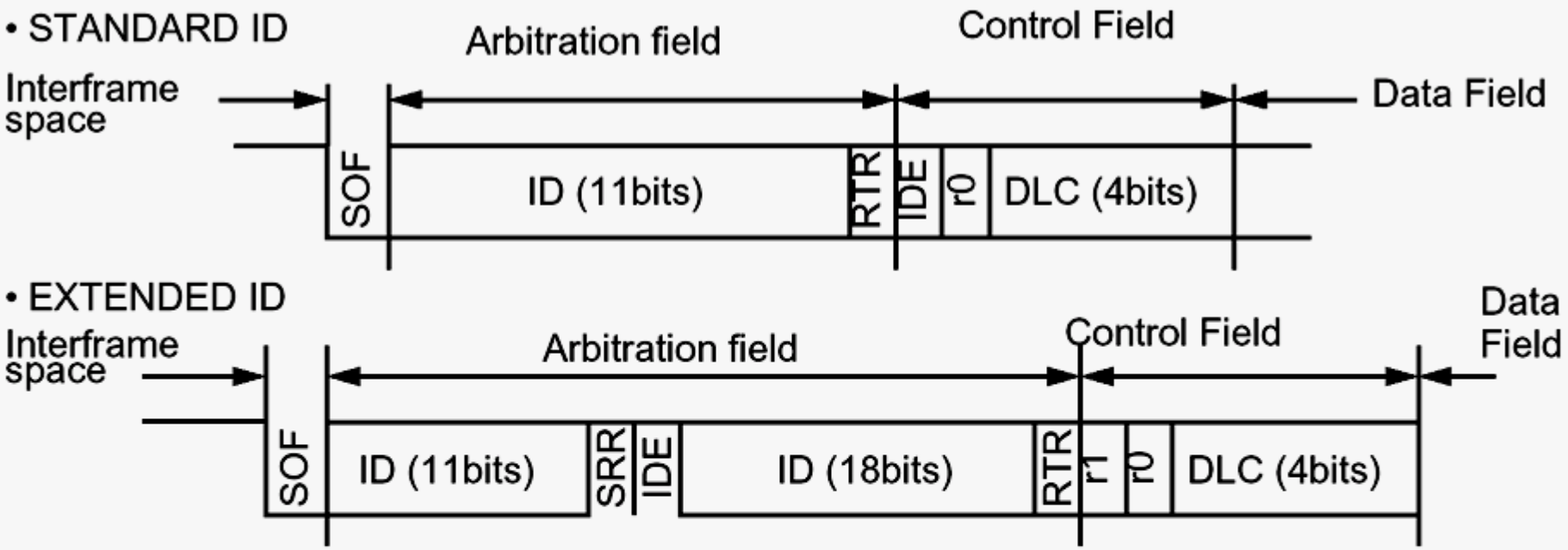
●Physical Layer of CAN





§5.3 网络通讯/ Network and Communication

CAN Frame Format Comparison 2.0 A and 2.0B



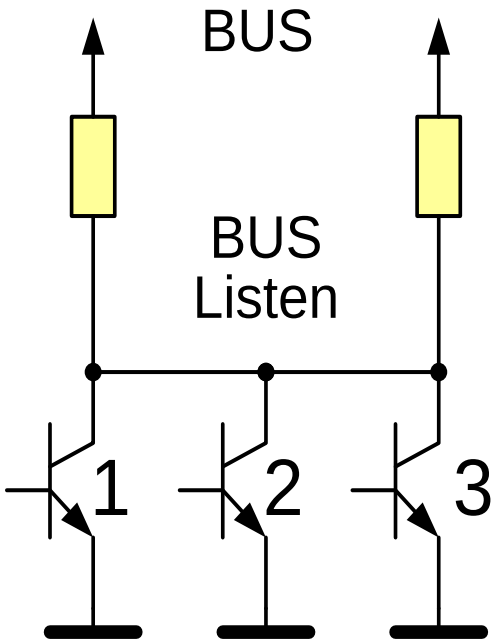
SOF: Start Of Frame
ID: Identifier
RTR: Remote Transmission Request
r1, r0: reserved bits
DLC: Data Length Code
IDE: Std or Extd. ID (CAN 2.0 B only)
SRR: Replacement of RTR on Std. format (CAN 2.0B)

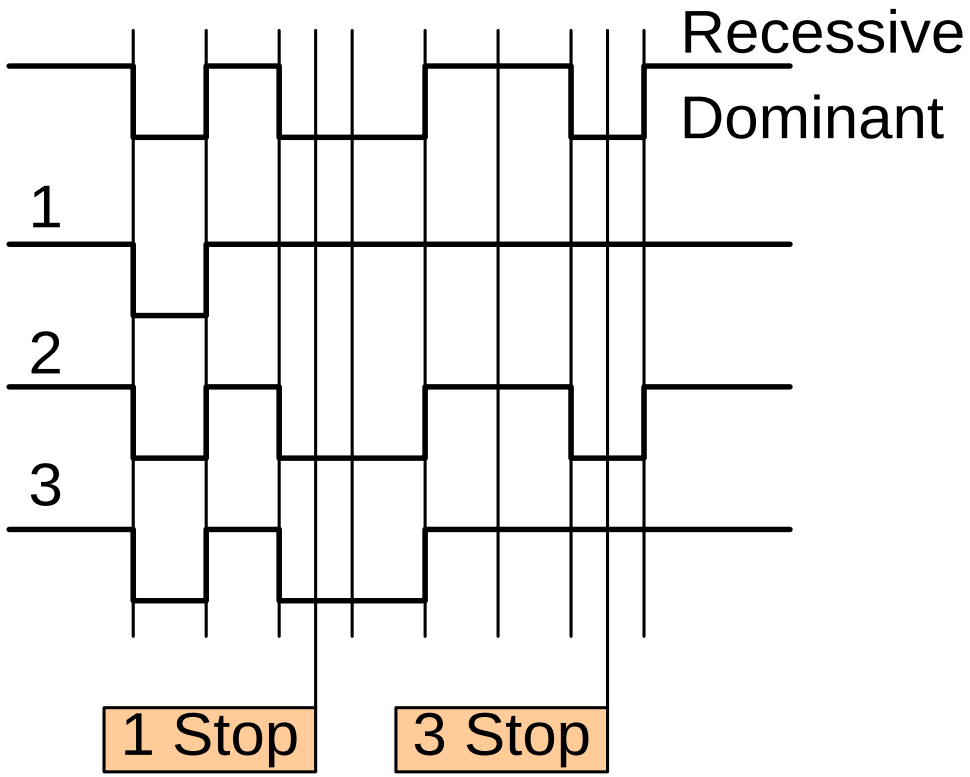


§5.3 网络通讯/ Network and Communication

●Bit to Bit Arbitration

- ◆Listen Before Send Out
- ◆Compare the Bus Level to Itself Bit Level
- ◆If Different then Stop Transfer

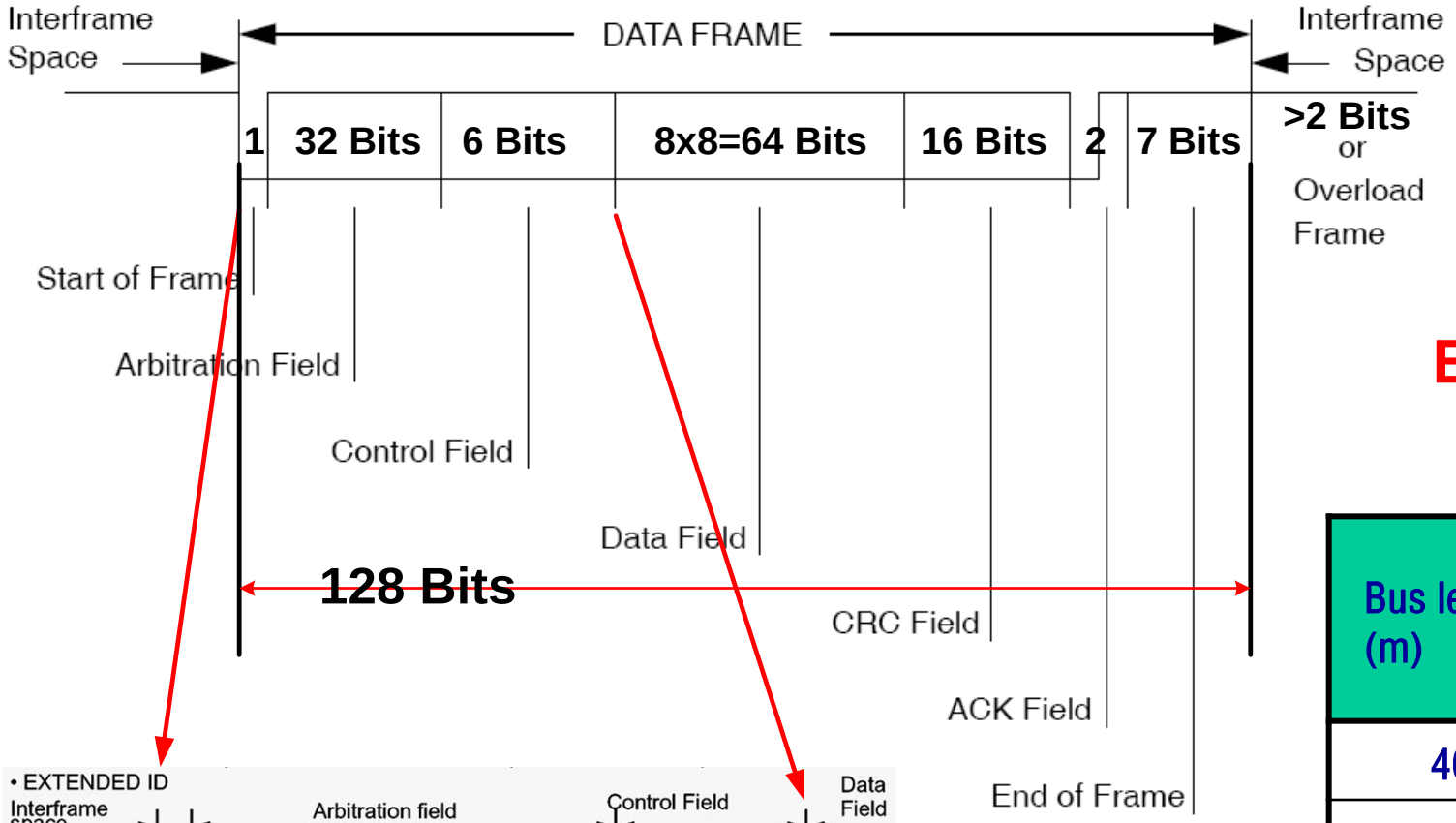


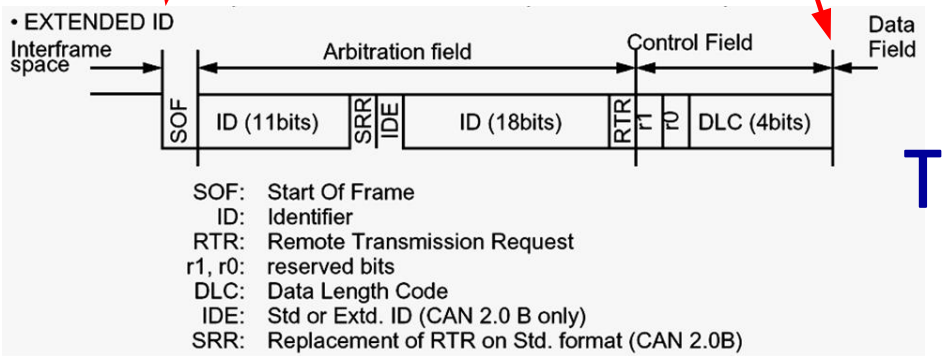


Node 1 = 1
Node 2 = 0
Node 3 = 1
Bus = 0



§5.3 网络通讯/ Network and Communication





Efficiency <50% !

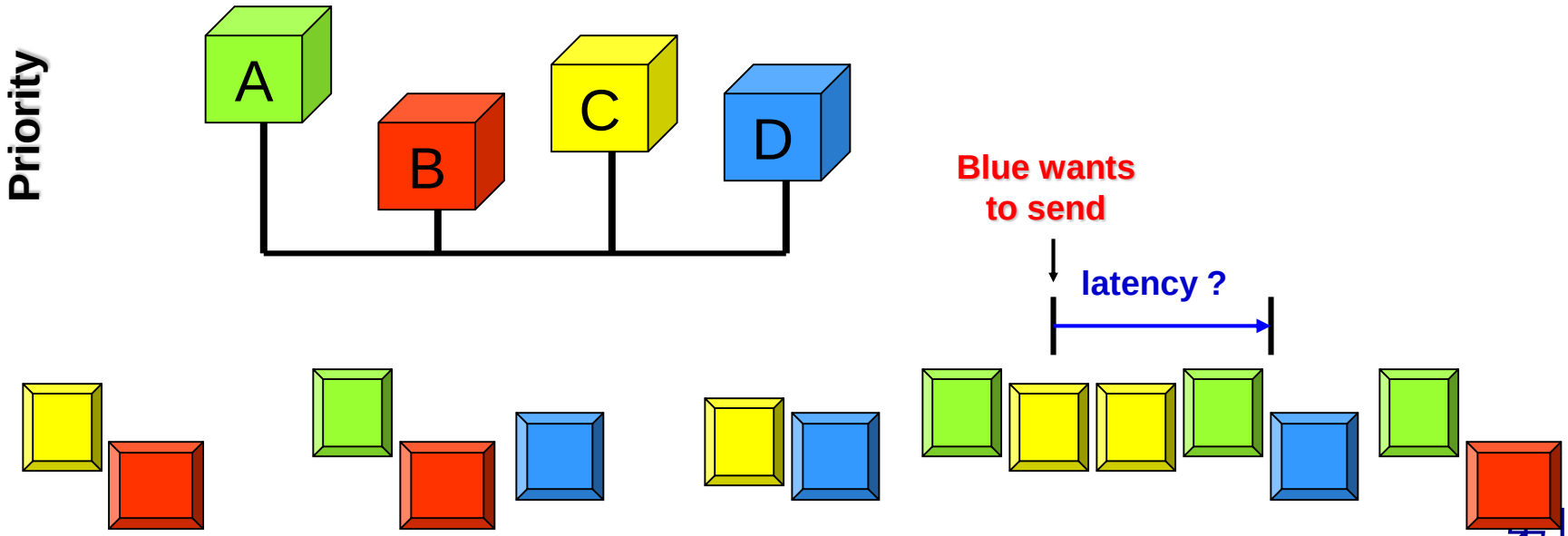
Transmit Time Calculation

Bus length (m)	Max. bit rate (b/s)
40	1M
100	500k
200	250k
500	125k
6 km	10k



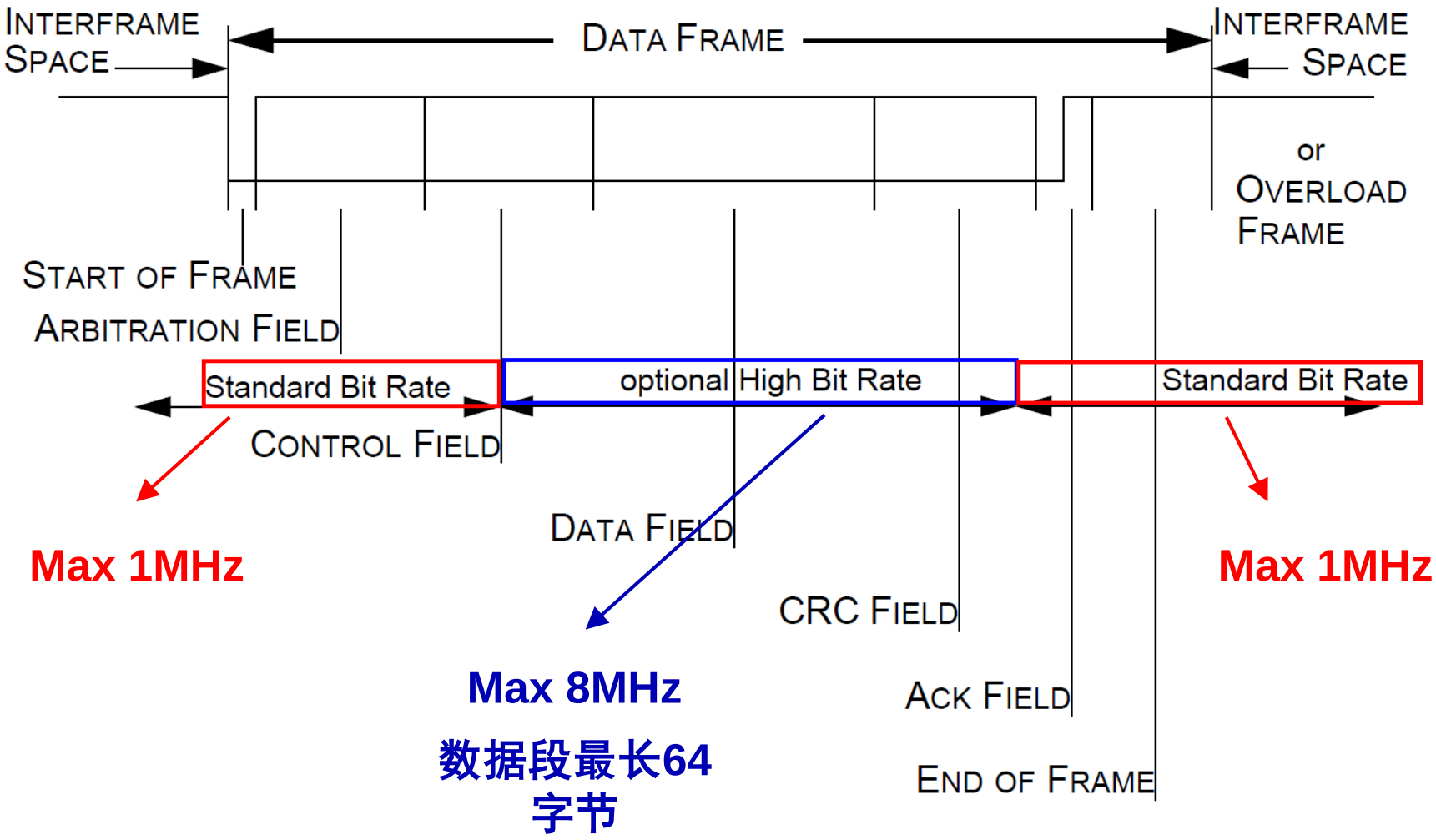
§5.3 网络通讯/ Network and Communication

- Disadvantage of CAN
  - ◆ Under the bitwise arbitration of CAN, the access may be delayed, if some other message is already in the process of transmission or if another message with higher priority also competes for the bus.
  - ◆Even the message with the highest priority may experience a small latency.
  - ◆The lower the priority of a message is, the higher the latency jitter for the media access





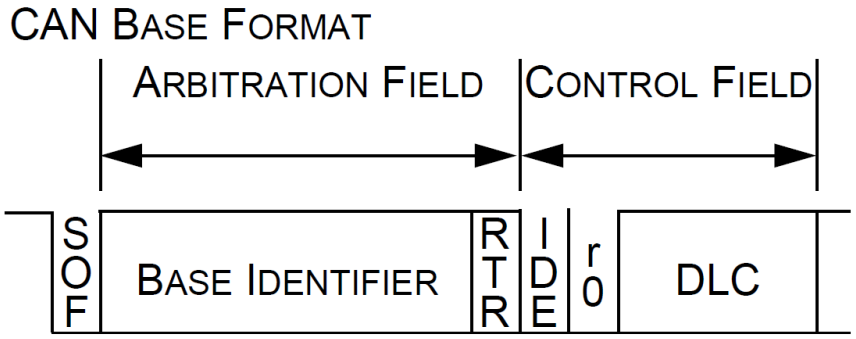
CAN_FD: CAN with Flexible Data-rate



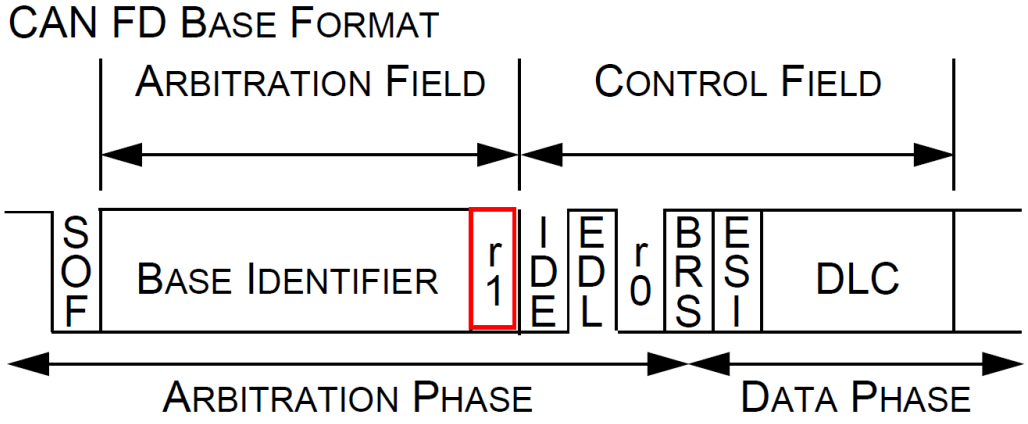


The Next Powertrain Network: CAN_FD vs Flexray

CAN_FD向下兼容传统CAN的硬件，兼容ISO-11898-1，CAN_FD扩展了传统CAN的性能：数量段和波特率提高。

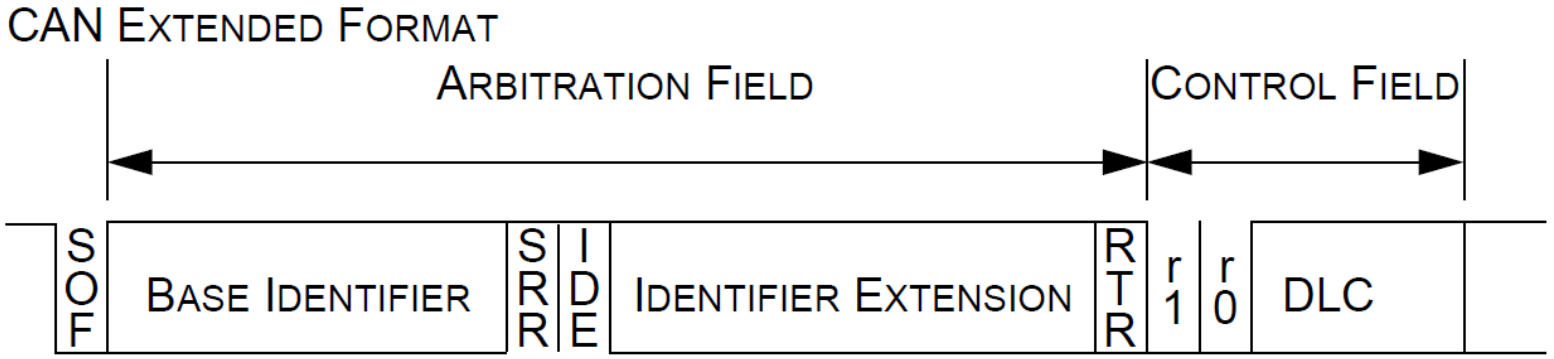


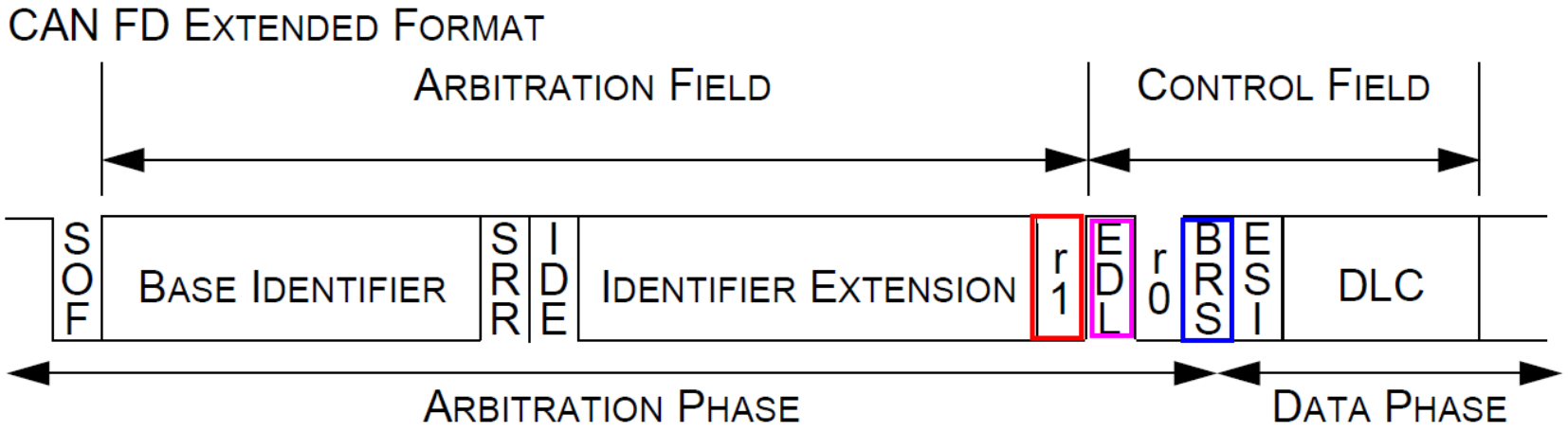
- Data Frame In CAN Base Format
- Data Frame In CAN Extended Format
- Data Frame In CAN FD Base Format
- Data Frame In CAN FD Extended Format





扩展帧格式


CAN EXTENDED FORMAT
ARBITRATION FIELD
CONTROL FIELD
SOF
BASE IDENTIFIER
SRR
IDE
IDENTIFIER EXTENSION
RTR
r1
r0
DLC


CAN FD EXTENDED FORMAT
ARBITRATION FIELD
CONTROL FIELD
SOF
BASE IDENTIFIER
SRR
IDE
IDENTIFIER EXTENSION
r1
EDL
r0
BRS
ESI
DLC
ARBITRATION PHASE
DATA PHASE



§5.3 网络通讯/ Network and Communication

DLC的定义

	Number of Data Bytes	Data Length Code			
		DLC3	DLC2	DLC1	DLC0
Codes in CAN and CAN FD Format	0	0	0	0	0
	1	0	0	0	1
	2	0	0	1	0
	3	0	0	1	1
	4	0	1	0	0
	5	0	1	0	1
	6	0	1	1	0
	7	0	1	1	1
CAN Format	8	1	0/1	0/1	0/1
Codes in CAN FD Format	8	1	0	0	0
	12	1	0	0	1
	16	1	0	1	0
	20	1	0	1	1
	24	1	1	0	0
	32	1	1	0	1
	48	1	1	1	0
	64	1	1	1	1



§5.3 网络通讯/ Network and Communication

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§5.3 网络通讯/ Network and Communication

- Background of TTP
  - The Time Triggered Protocol TTP was designed at the Technical University of Vienna to meet the stringent requirements for distributed fault tolerant control systems.
  - Its medium access strategy is based on a static time division multiple access TDMA scheme.
  - The total channel capacity is divided into a number of time slots that are statically assigned to the nodes.
  - This a priori knowledge is used for the efficient implementation of several protocol services.



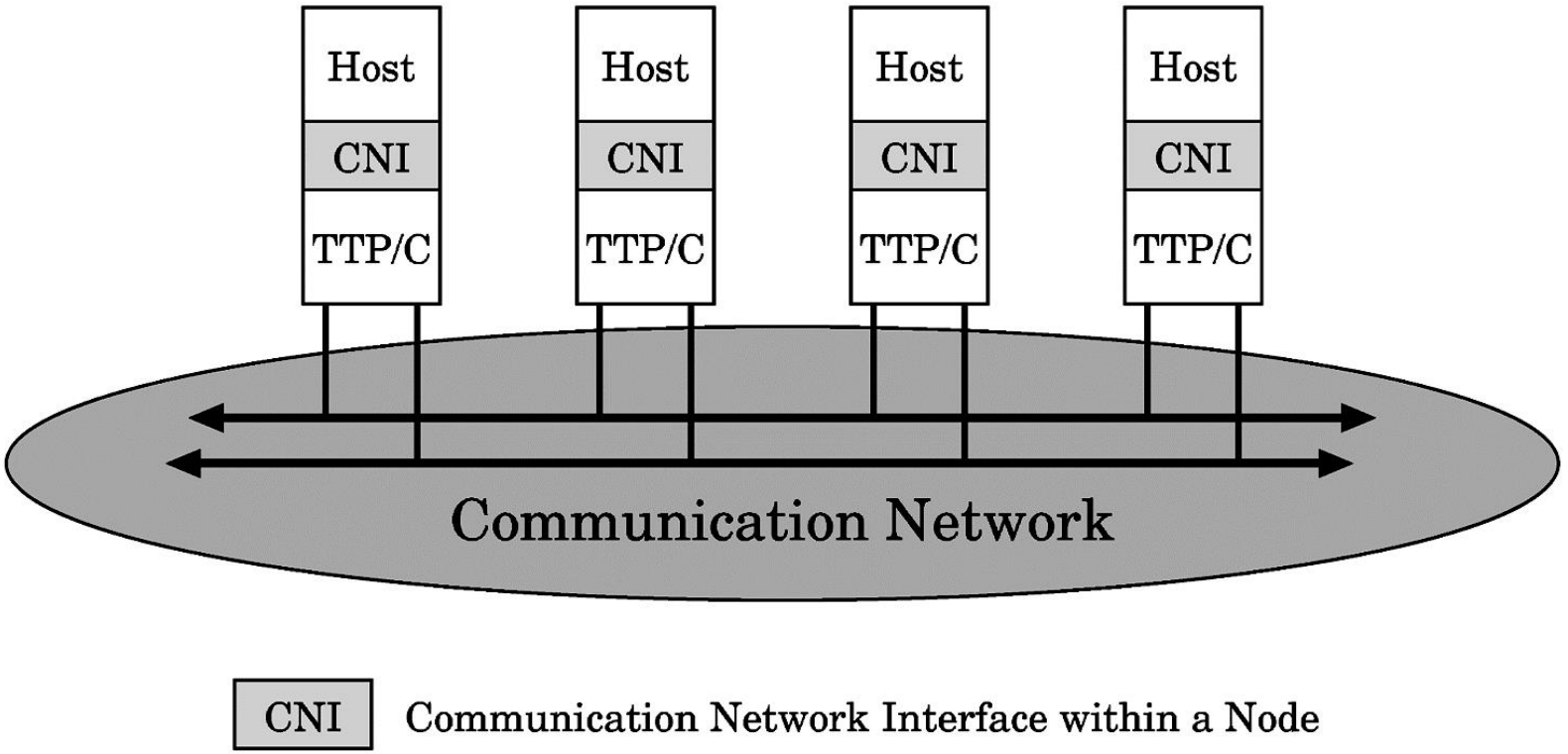
§5.3 网络通讯/ Network and Communication

- TTP/C
  - The TTP/C protocol is the full version of the time triggered protocol that includes all above mentioned protocol services. Application areas are fault tolerant distributed hard real–time systems e.g., the SAE class C environment.
- TTP/A
  - The TTP/A protocol is a low–cost member of the TTP protocol family that is intended for non fault tolerant field bus applications. TTP/A detailed description is contained in [Kop95b].



§5.3 网络通讯/ Network and Communication

●Double Channel, Line Topology



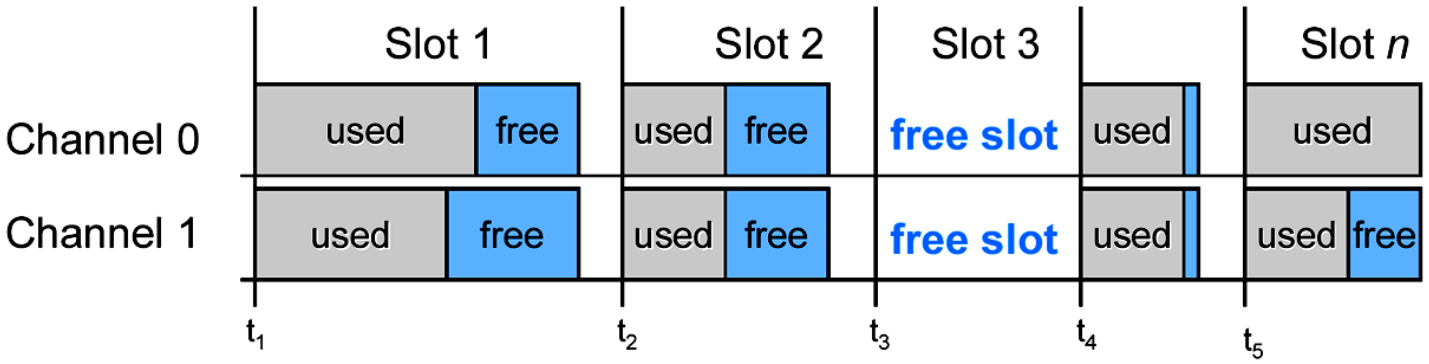
CNI Communication Network Interface within a Node



§5.3 网络通讯/ Network and Communication

- Time–Triggered Operation
  - System activities controlled by time
  - No dependence on events
  - Time available always and everywhere
  - TTP/C Transmission rates up to 25MHz

Time Division Multiple Access

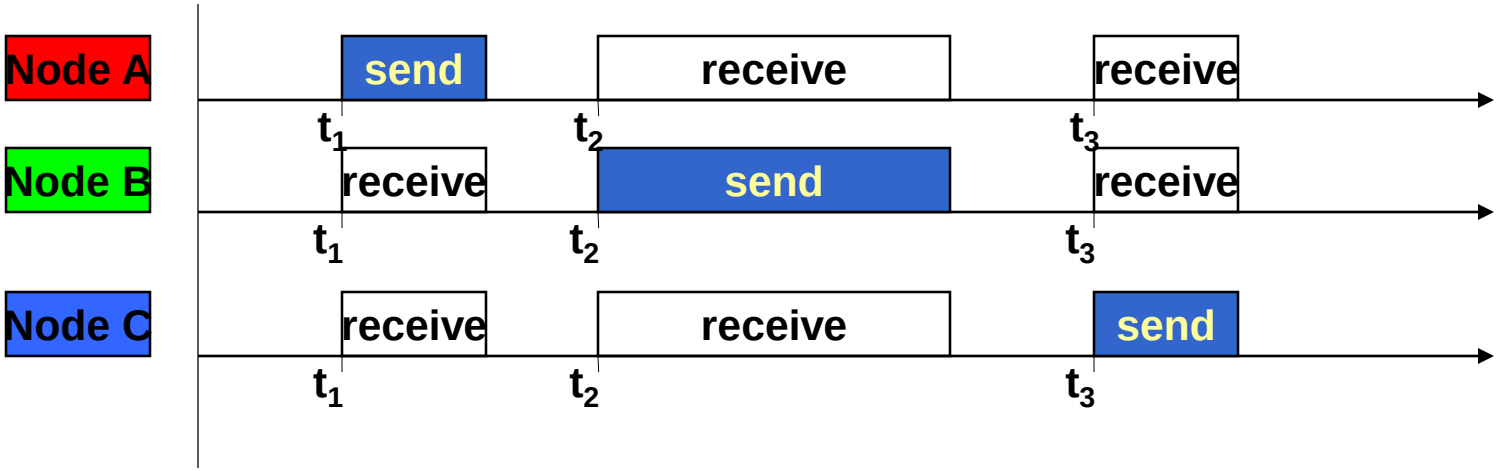






§5.3 网络通讯/ Network and Communication

- Characteristics of TTP/C Protocol
  - All nodes agree *a priori* on the points in time when send and receive operations are carried out
  - TDMA media access is collision–free and immediate

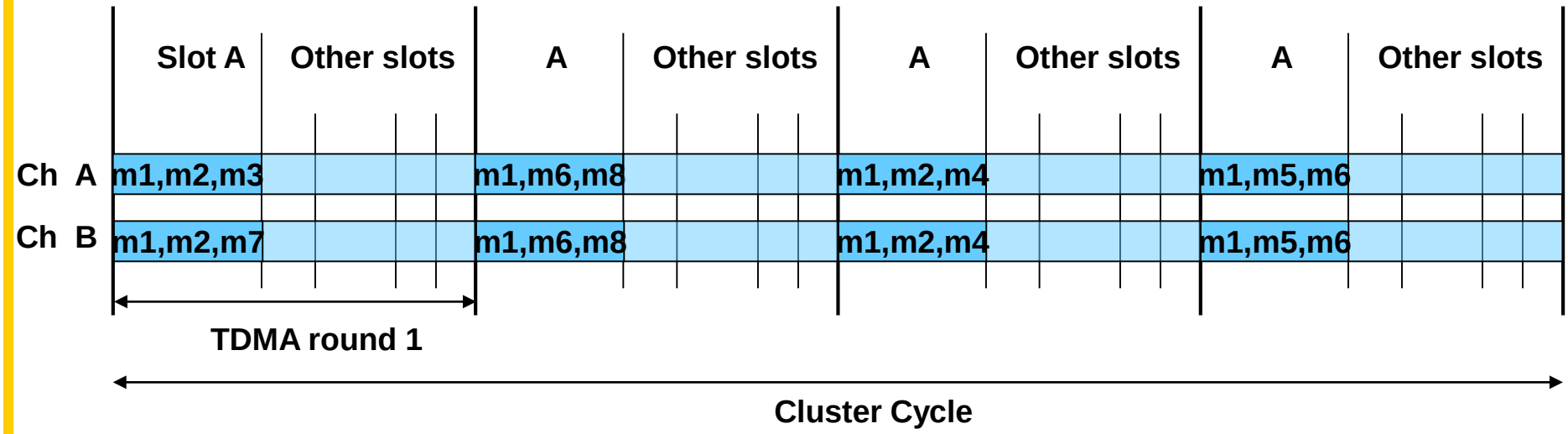


TDMA Protocol: Clock Synchronous

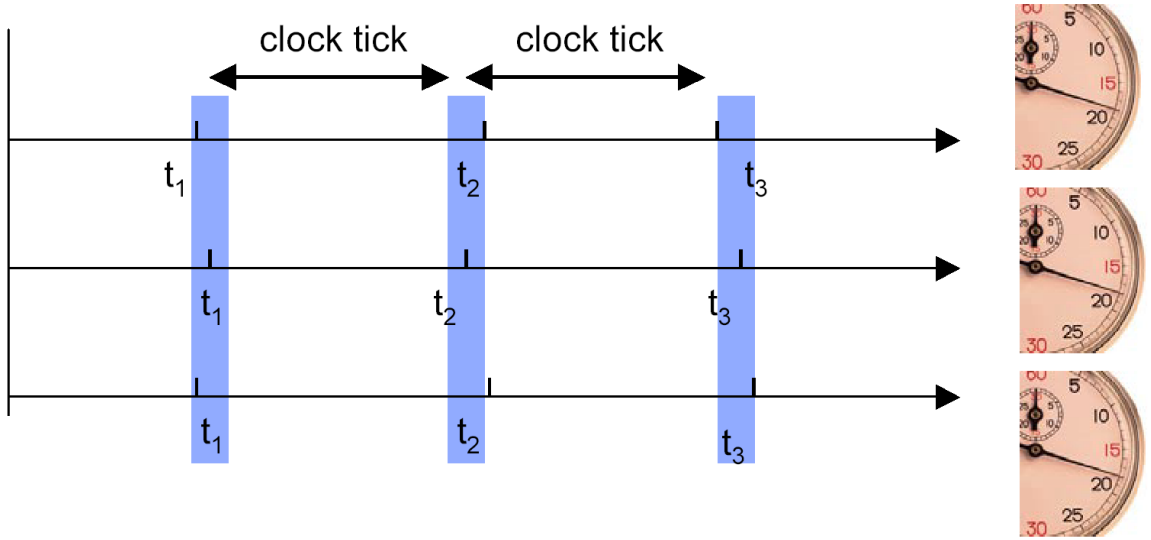


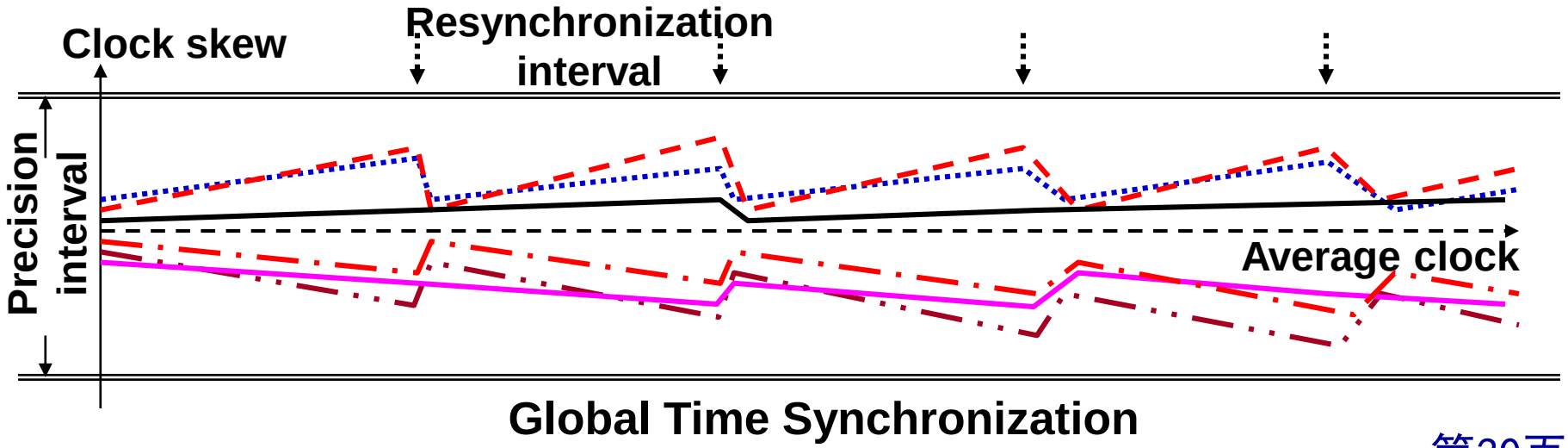
§5.3 网络通讯/ Network and Communication

- Messages, Slots, Rounds, Cycles
- Example: the cluster cycle has 4 TDMA rounds; Node A transmits :
  - round 1: m1, m2, m3, m7
  - round 2: m1, m6, m8
  - round 3: m1, m2, m4
  - round 4: m1, m5, m6
- m1 is updated every TDMA round, while m3 is only updated once every cluster cycle.



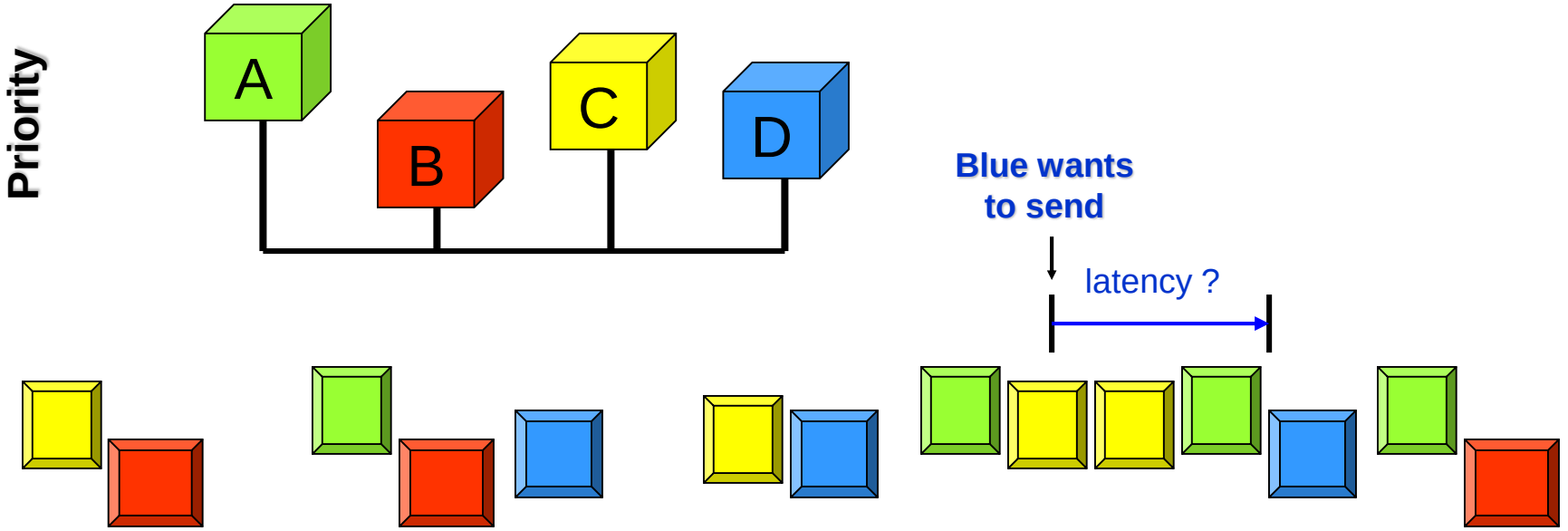
●Fault–Tolerant Global Time–Base





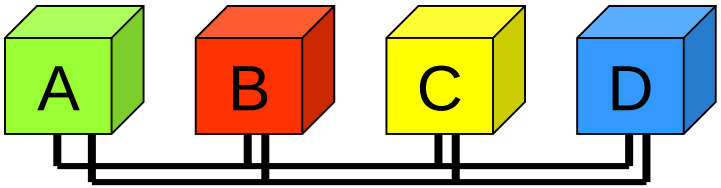


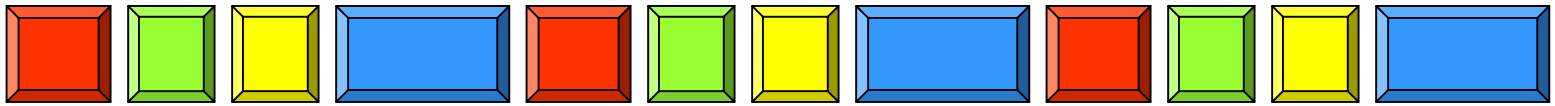
§5.3 网络通讯/ Network and Communication



Time Latency of CAN

CAN vs TTP/C





Real-Time Characteristics of TTP/C



§5.3 网络通讯/ Network and Communication

- Introduction/简介
- Classifications of Bus System in Vehicle
- Brief Overview of CAN (Controller Area Network)
- Introduction of TTP/C (Time Triggered Protocol / Class C)
- FlexRay Protocol
- TTCAN Protocol
- Applications of TTCAN

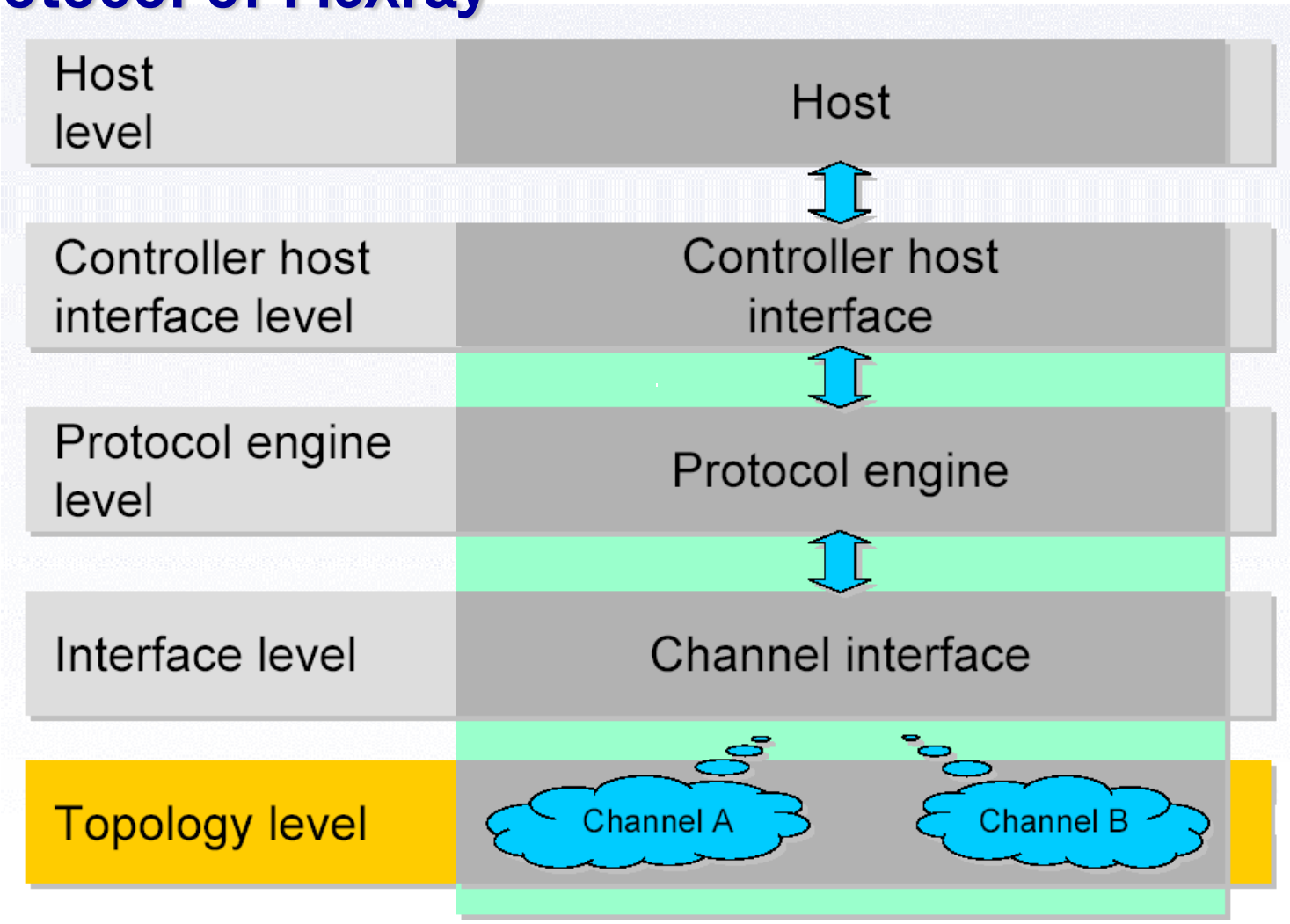


§5.3 网络通讯/ Network and Communication

- FlexRay：
  - The significant difference between Flexray and Byteflight is that the later bases on optical fiber, which has lower EMI but higher cost.
  - Both have high transmission rate up to 10~25MHz ;
  - Both Based on FTDMA with synchronous and asynchronous message;
  - Both can be applied in safety critical system, e.g. “X–by wire”.
  - Contrast to the MOST, Both of Flexray and Byteflight are oriented to real–time control systems of vehicles.



●Protocol of Flexray

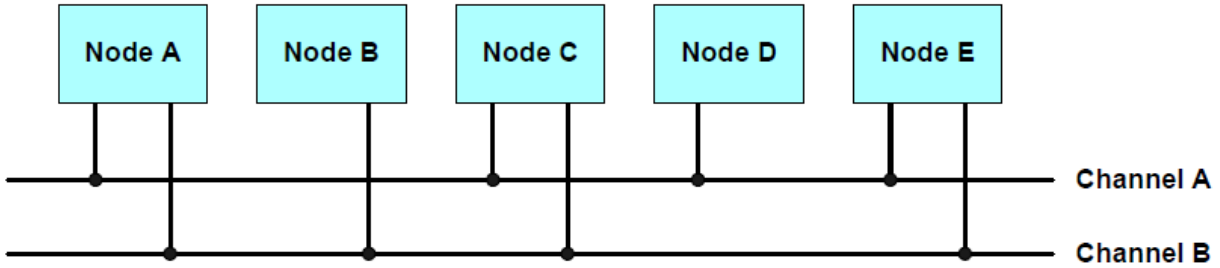


FlexRay Architecture

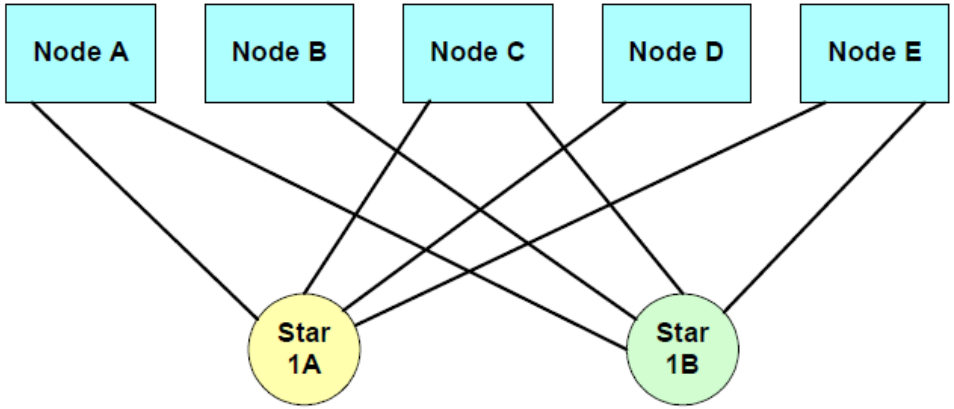


§5.3 网络通讯/ Network and Communication

●Protocol of Flexray



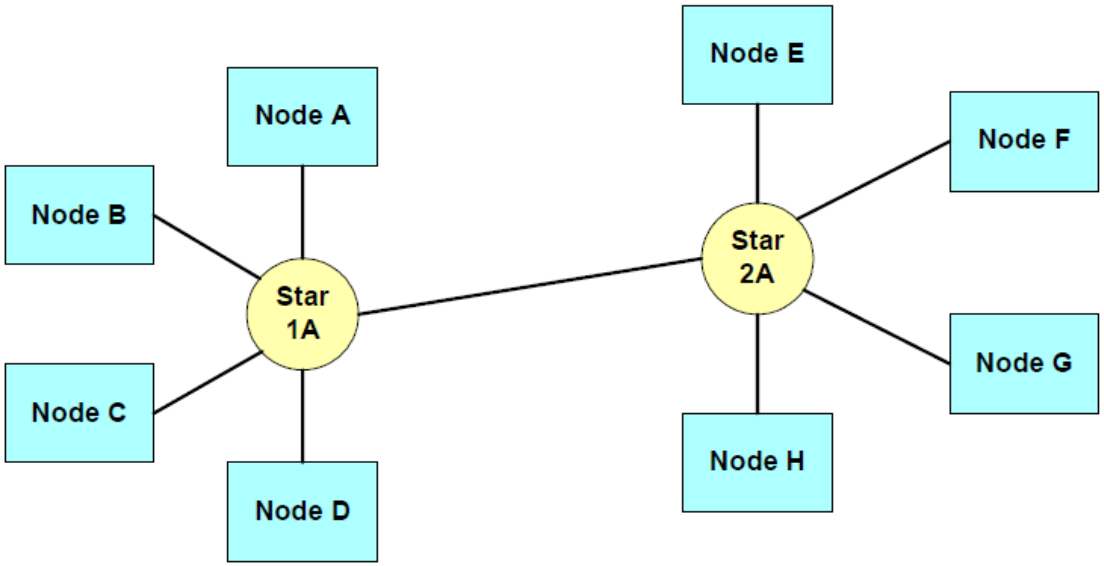
Line Topology



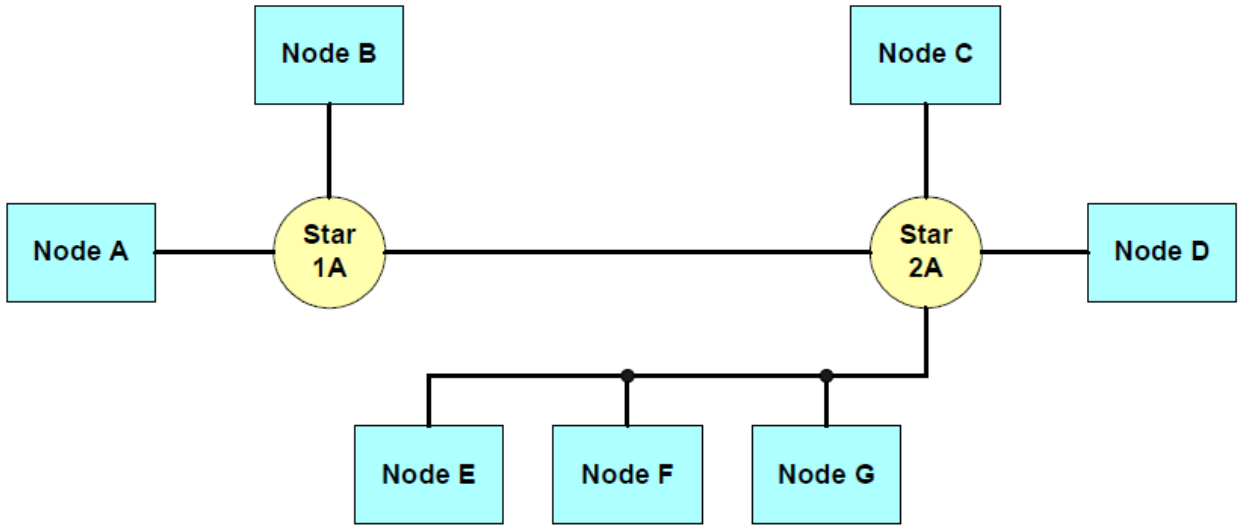
Star Topology



§5.3 网络通讯/ Network and Communication

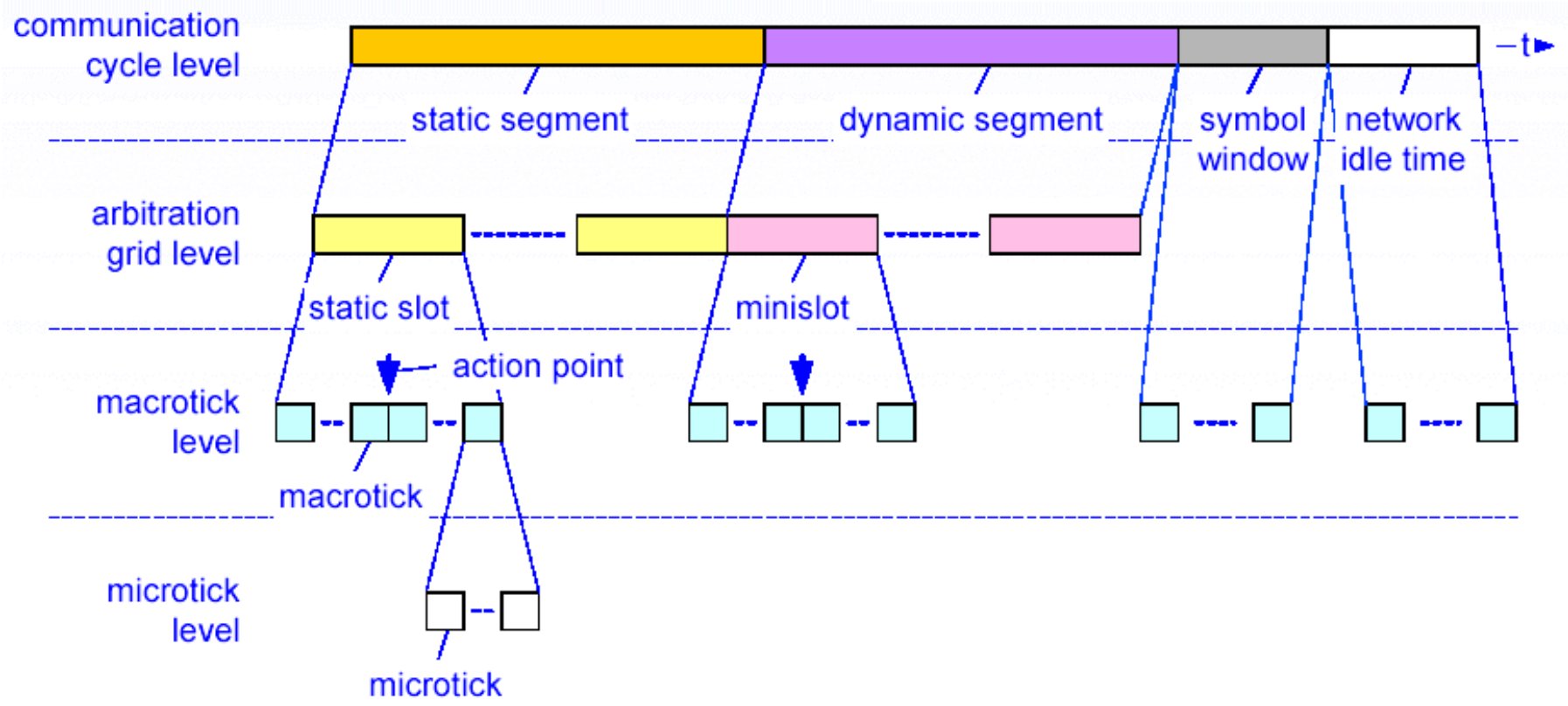


Star Topology





§5.3 网络通讯/ Network and Communication

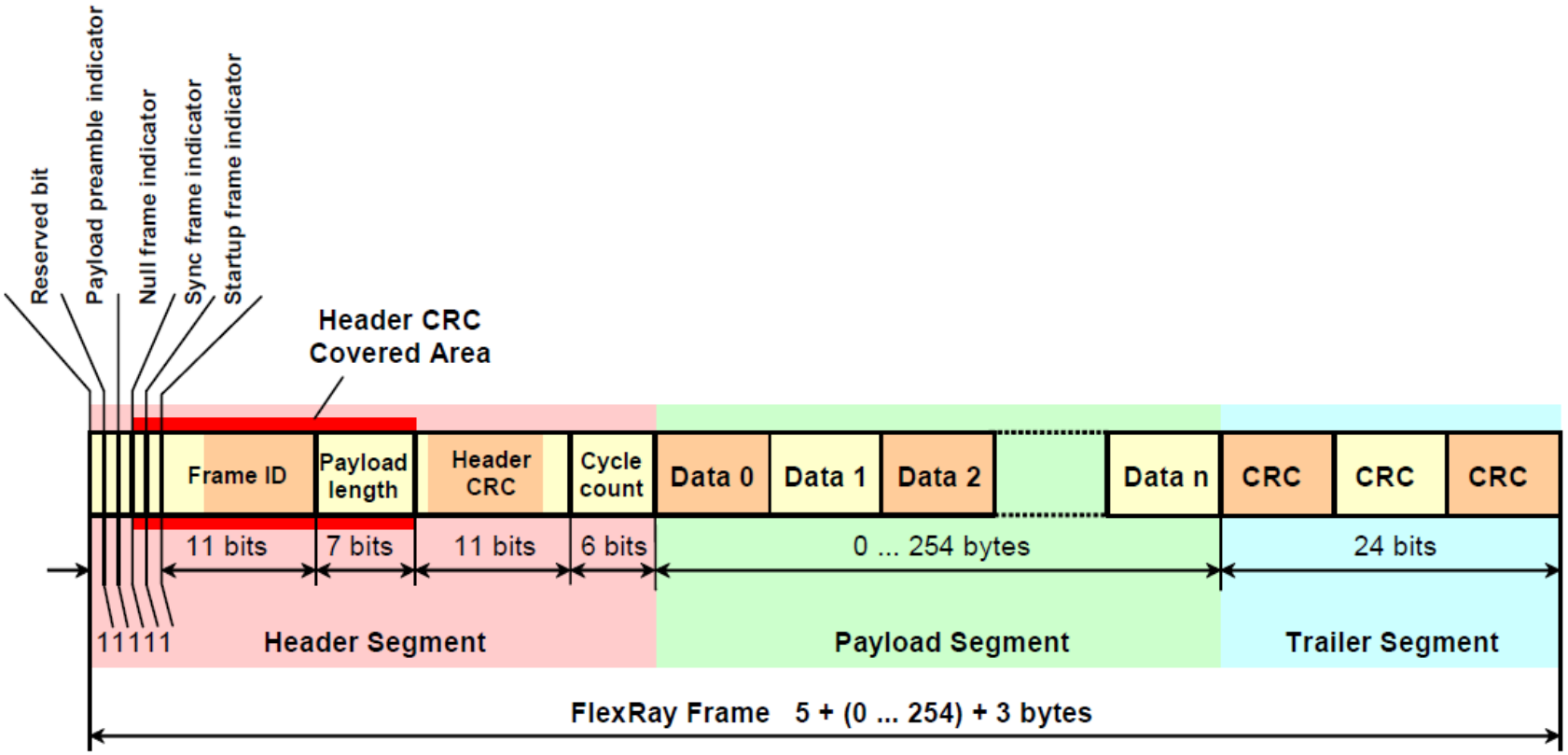


FlexRay Protocol Timing



§5.3 网络通讯/ Network and Communication

- Frame Format (Final) released in 2005



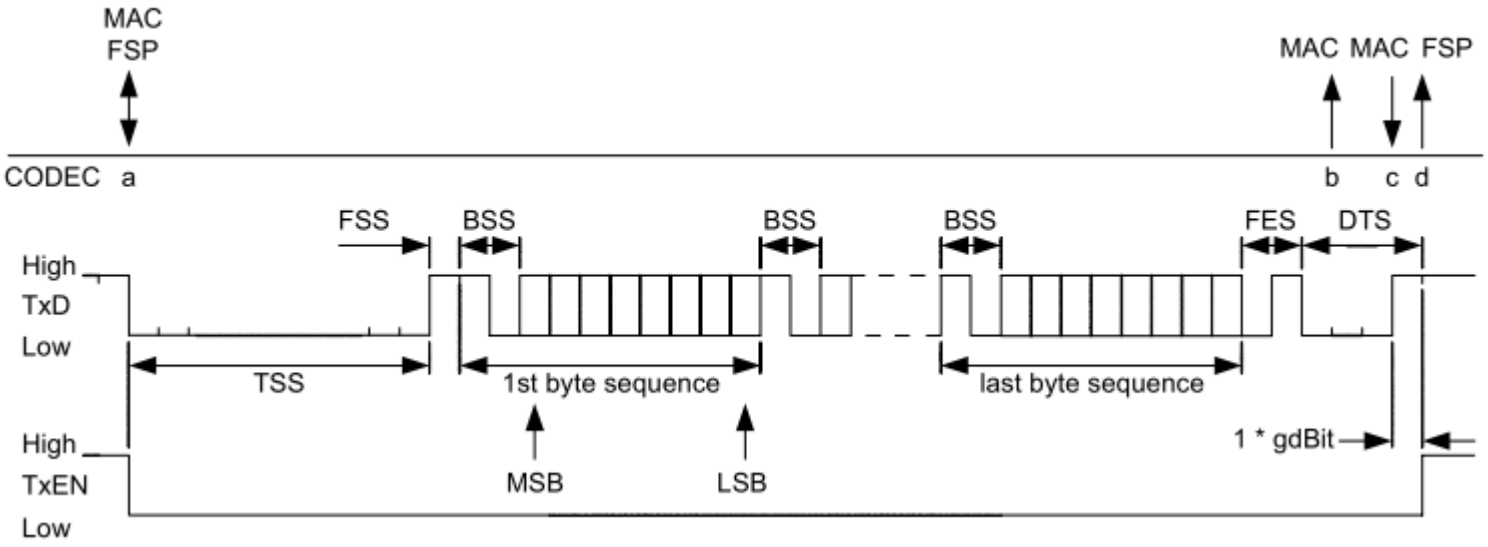
Flexray-message format with Frame ID

Flexray简介

Frame bit stream assembly

Dynamic Segment: 10MHz波特率，但是每个字节传输10位；

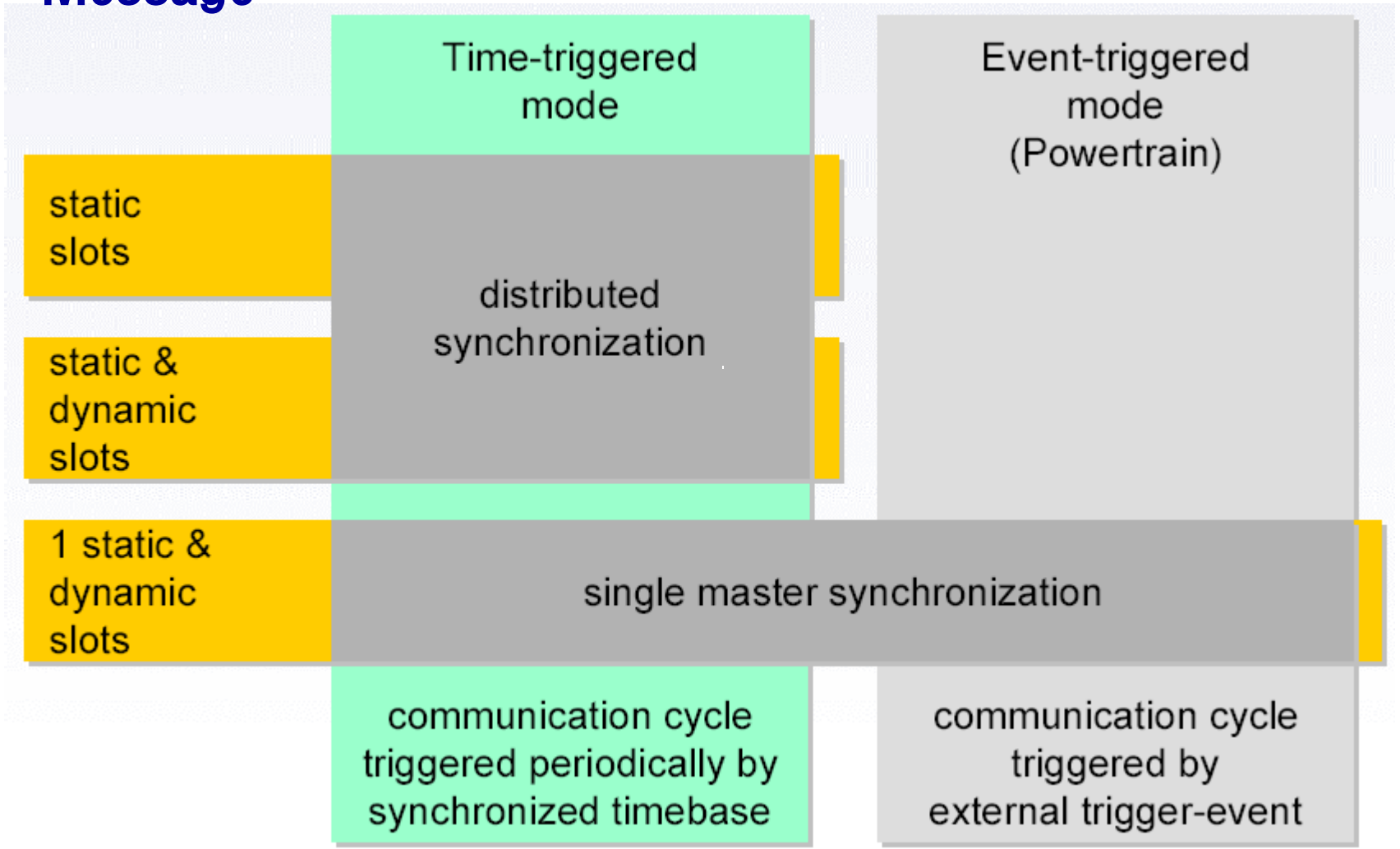
FSP→TSS→FSS→BSS→Data1→BSS→Data2→…BSS→DataN→FES→DTS→FSP





§5.3 网络通讯/ Network and Communication

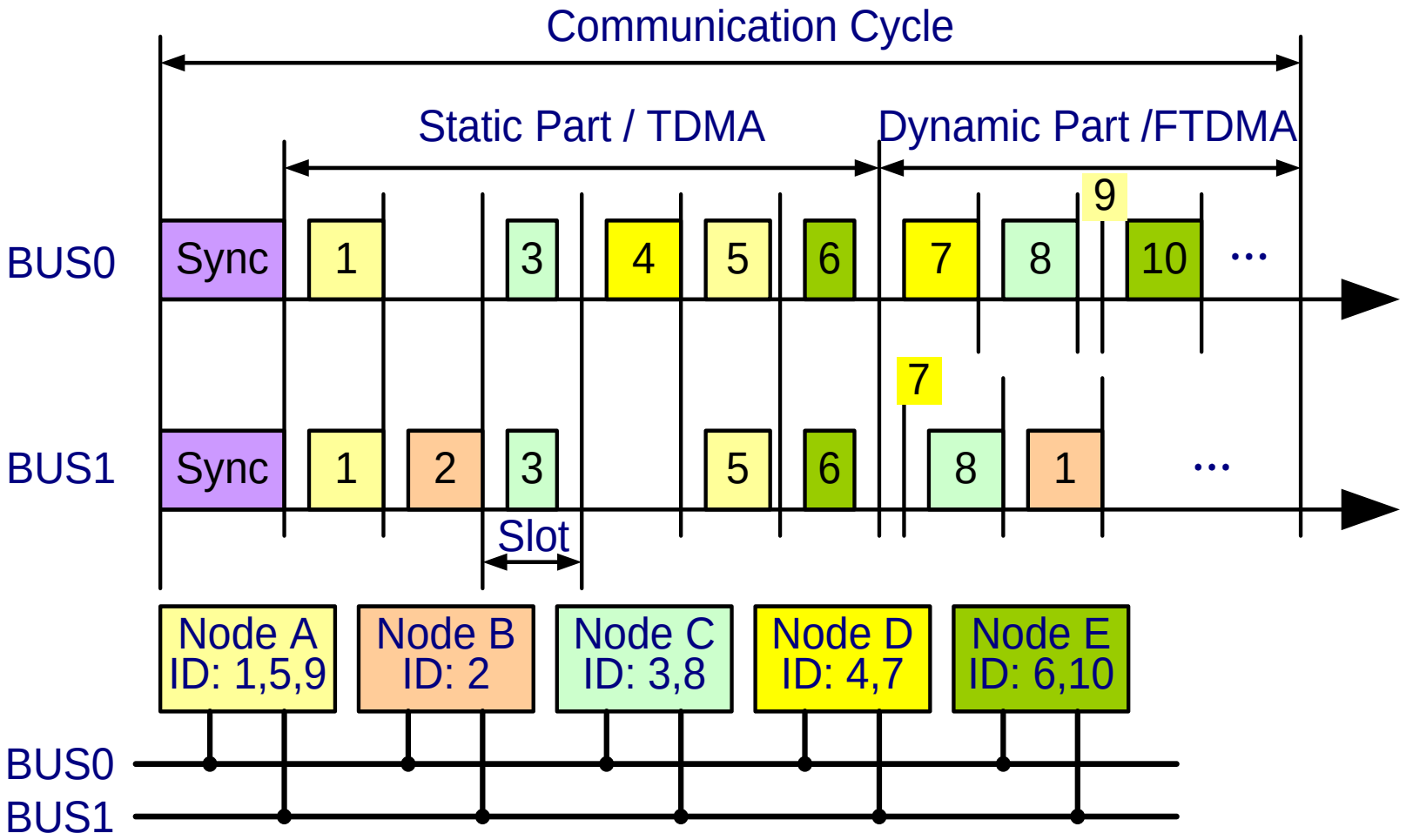
- Message



FlexRay Application Requirements



§5.3 网络通讯/ Network and Communication

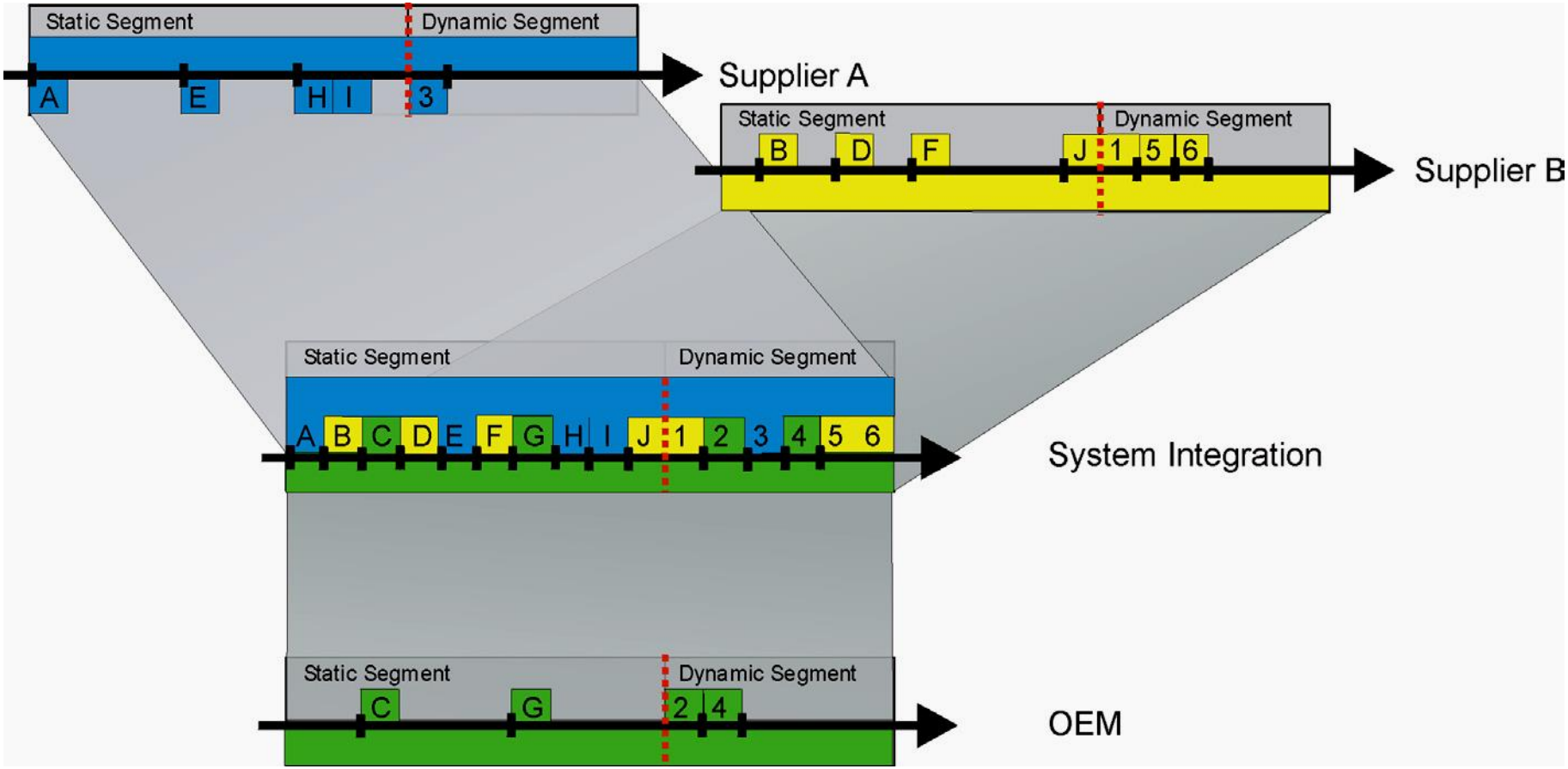


FlexRay Communication Modes:
Synchronous and Asynchronous



§5.3 网络通讯/ Network and Communication

- System integration with different parts with FlexRay

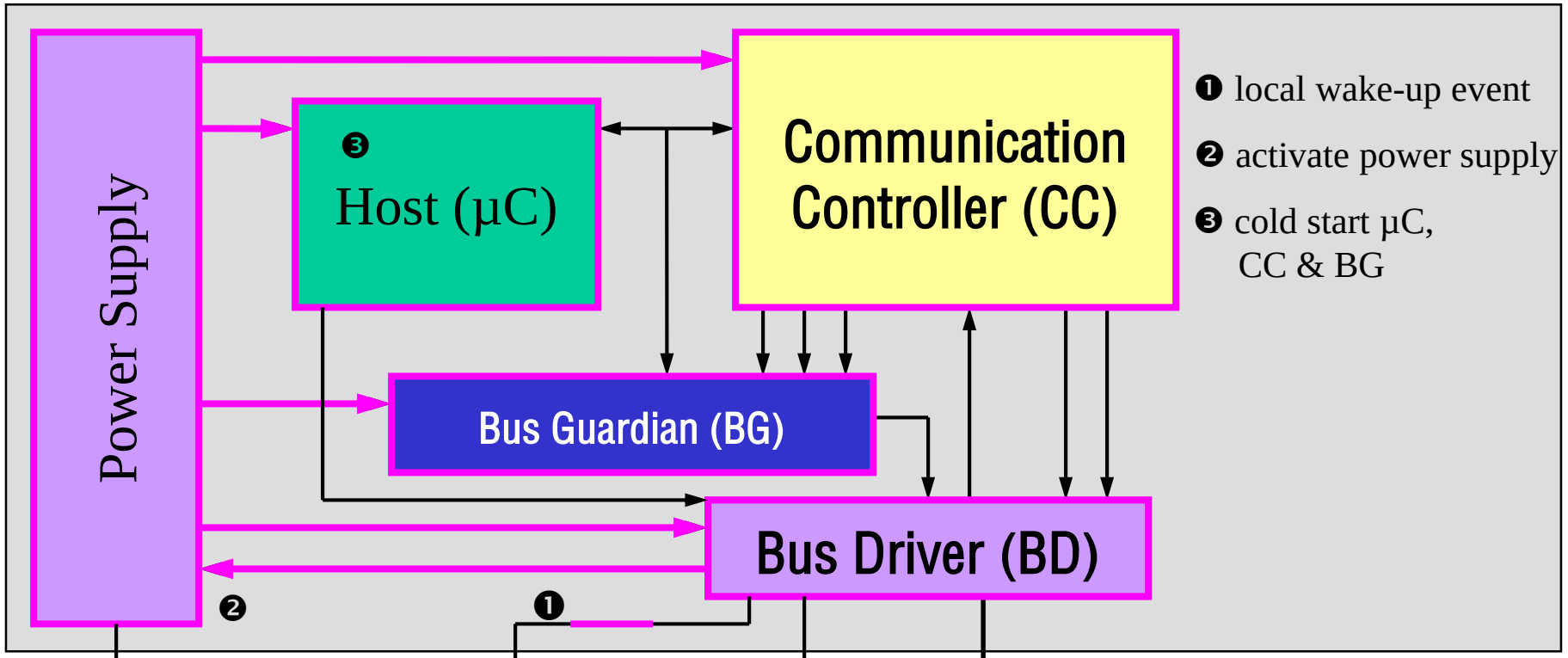


Application in Car: System Integration



§5.3 网络通讯/ Network and Communication

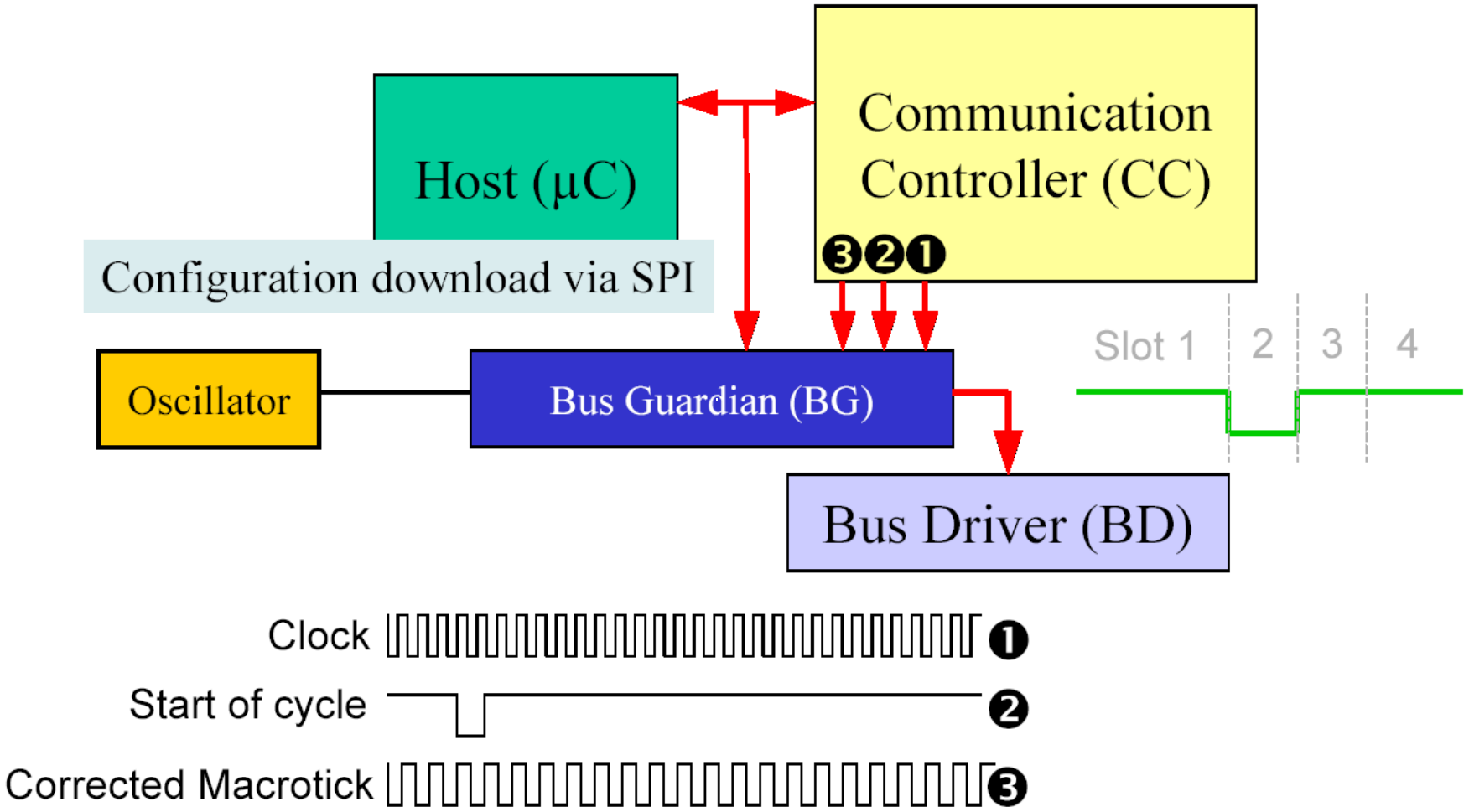
●The Physical Layer



Framework of the Physical Layer of Flexray



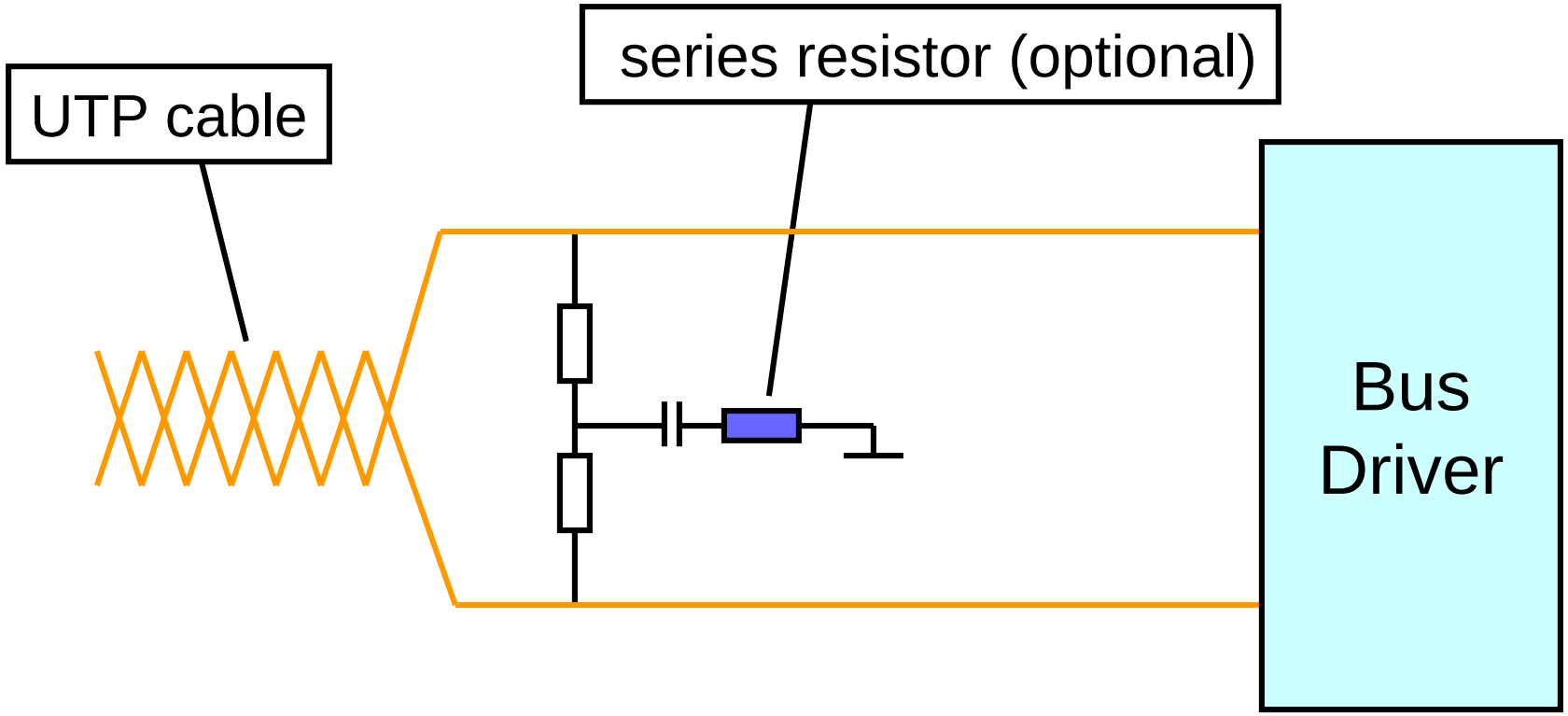
§5.3 网络通讯/ Network and Communication



The Physical Layer of Flexray



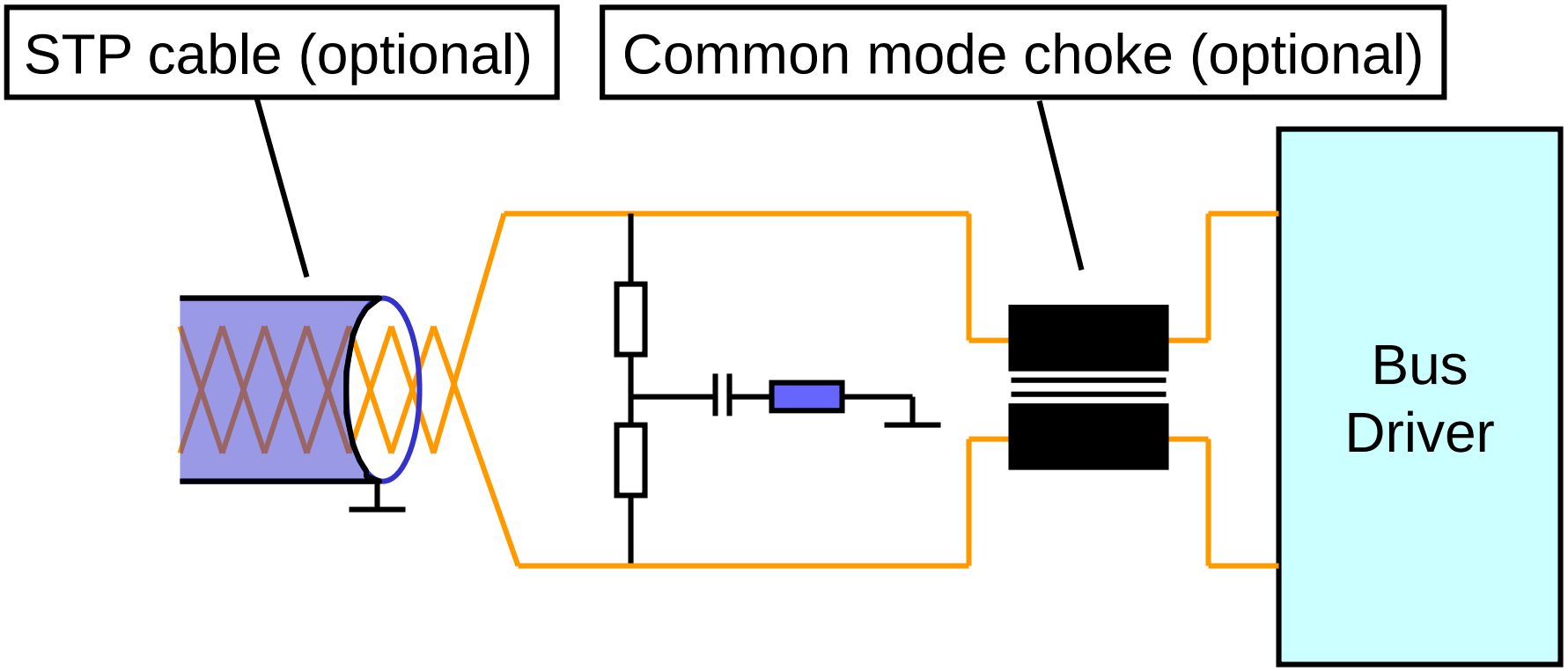
- EMI Problem are Considering



Cable Termination of Flexray: EMI Reduction with UTP Cable and Series Resistor



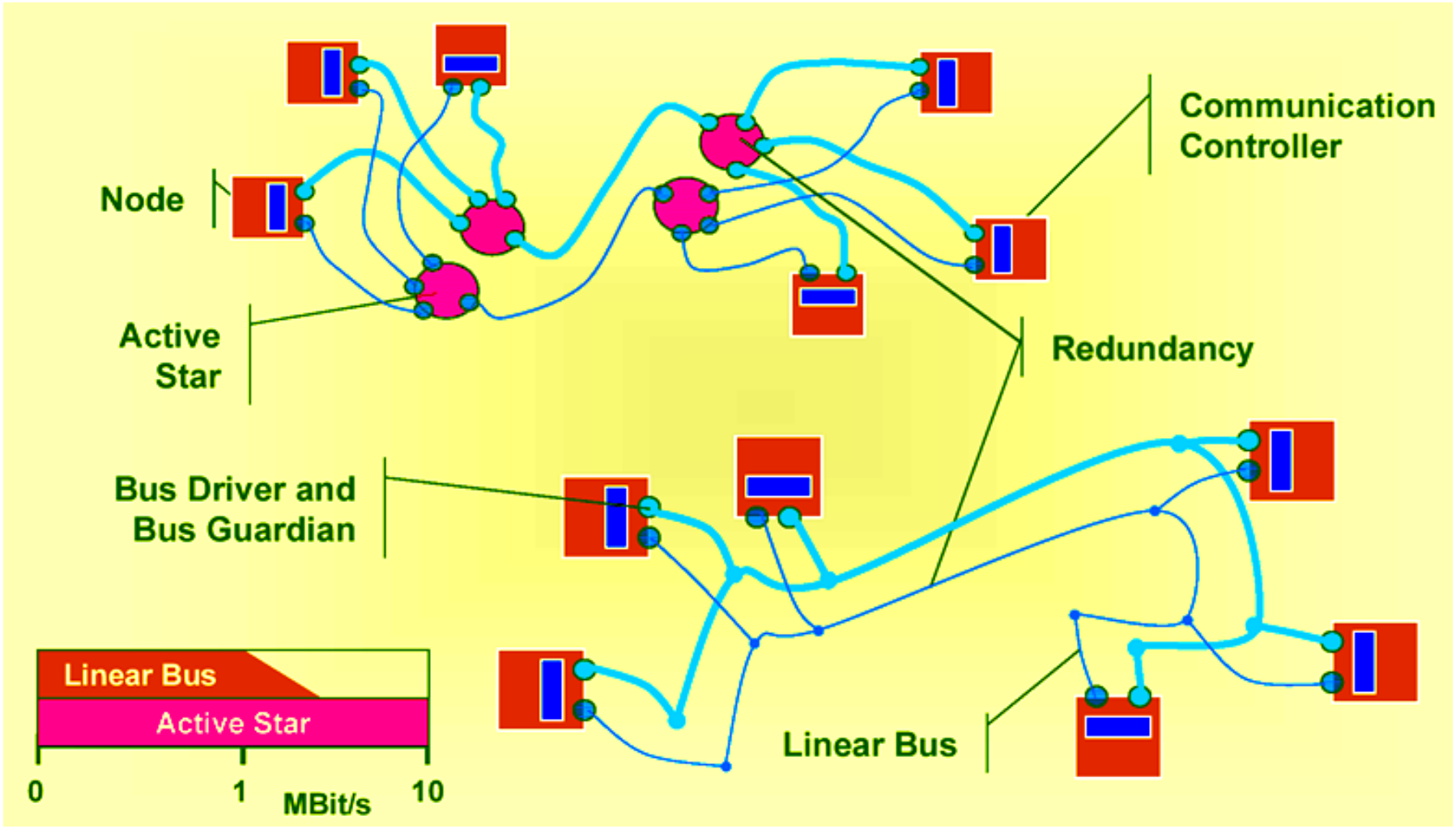
§5.3 网络通讯/ Network and Communication



Cable Termination of Flexray: EMI Reduction with Common Mode Choke and STP Cable



§5.3 网络通讯/ Network and Communication

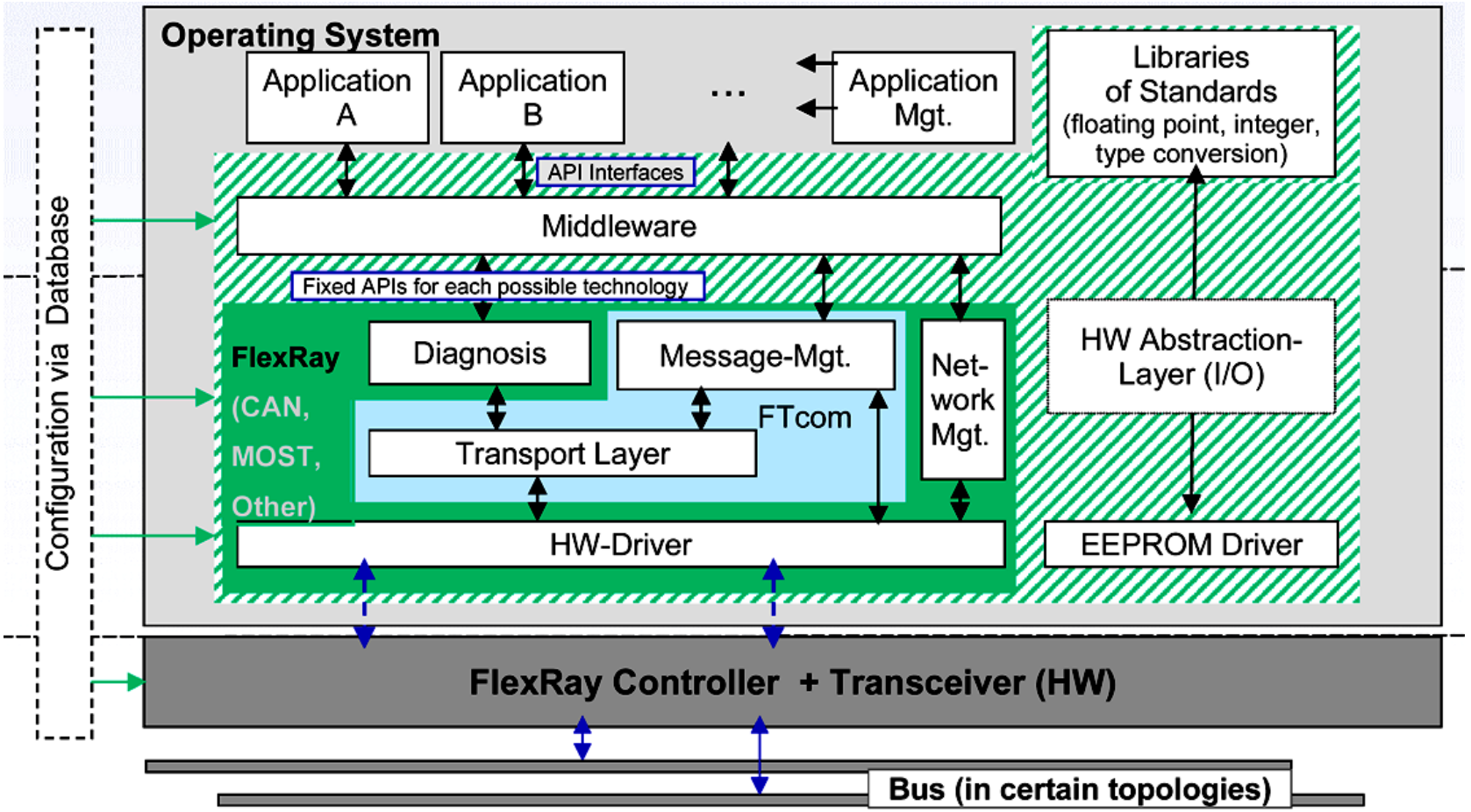


System Architecture FlexRay Passive Bus and Active Star



§5.3 网络通讯/ Network and Communication

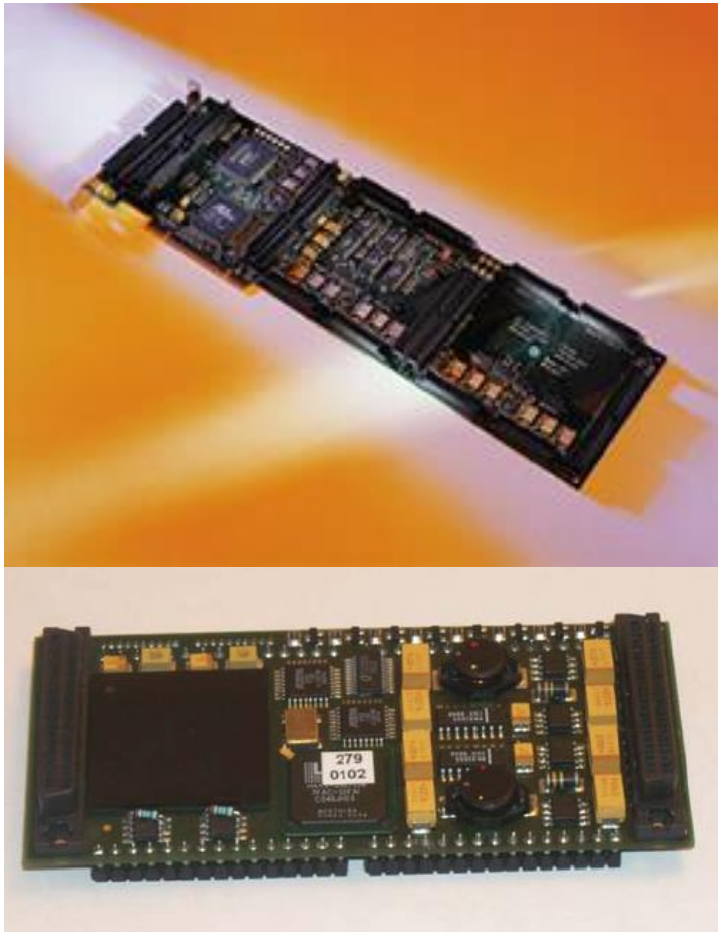
●Application and Development of Flexray



Standard-Structure model with FlexRay as underlying Communication Layer

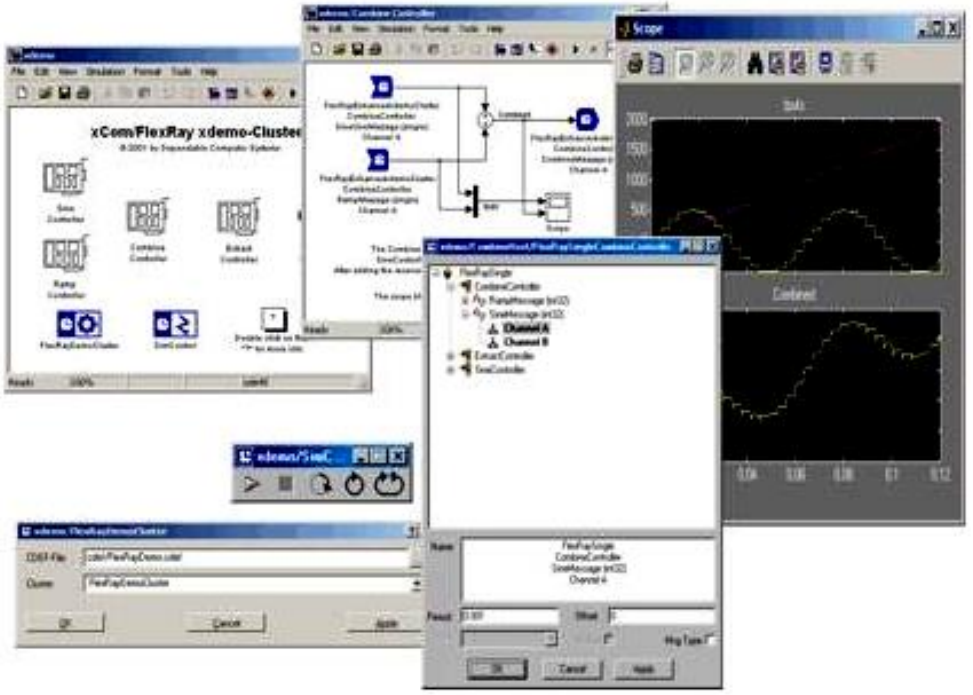


§5.3 网络通讯/ Network and Communication



xCom/FlexRay and xCom/TTP

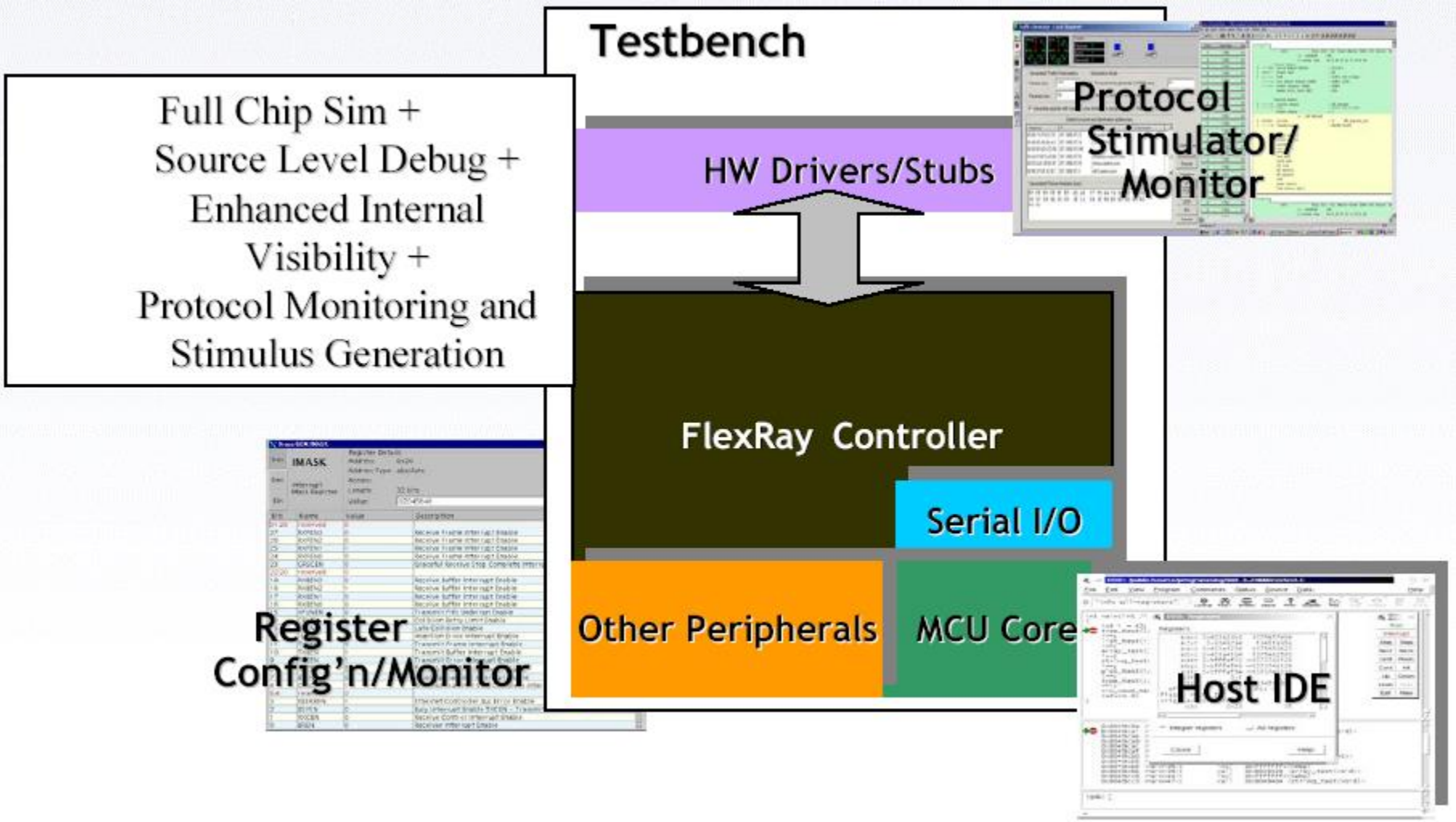
Time-triggered communication blocksets for use with MATLAB/Simulink



Application Tools of FlexRay: MATLAB xCOM Blockset



§5.3 网络通讯/ Network and Communication



Flexray Node Software Development Environment [Motorola]



§5.3 网络通讯/ Network and Communication

- Introduction/简介
- Classifications of Bus System in Vehicle
- Brief Overview of CAN (Controller Area Network)
- Introduction of TTP/C (Time Triggered Protocol / Class C)
- FlexRay Protocol
- TTCAN Protocol
- Applications of TTCAN



Event Trigger + Time Base Trigger

- The communication is based on the periodic transmission of a reference message by a time master.
- This allows to introduce a system wide global network time with high precision.
- Based on this time the different messages are assigned to time windows within a basic cycle.
- A big advantage of TTCAN compared to classic scheduled systems is the possibility to transmit event triggered messages in certain "arbitrating" time windows as well.
- These time windows, where normal arbitration takes place, allow the transmission of spontaneous messages.



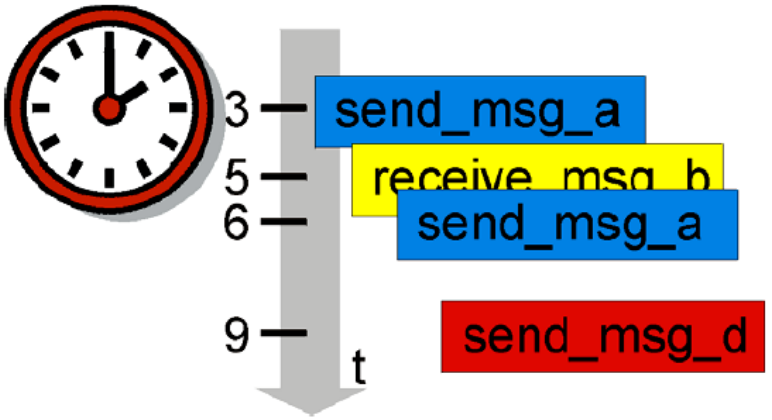
§5.3 网络通讯/ Network and Communication

Message A is sent if the system clock reaches 3 and 6 while message B is sent at 5.

Global Time Table→ The New Transmission is Deterministic and Predictable → TTCAN

The Time Base must be mapped into each node within the network. The remain parts of TTCAN is the same as general CAN.

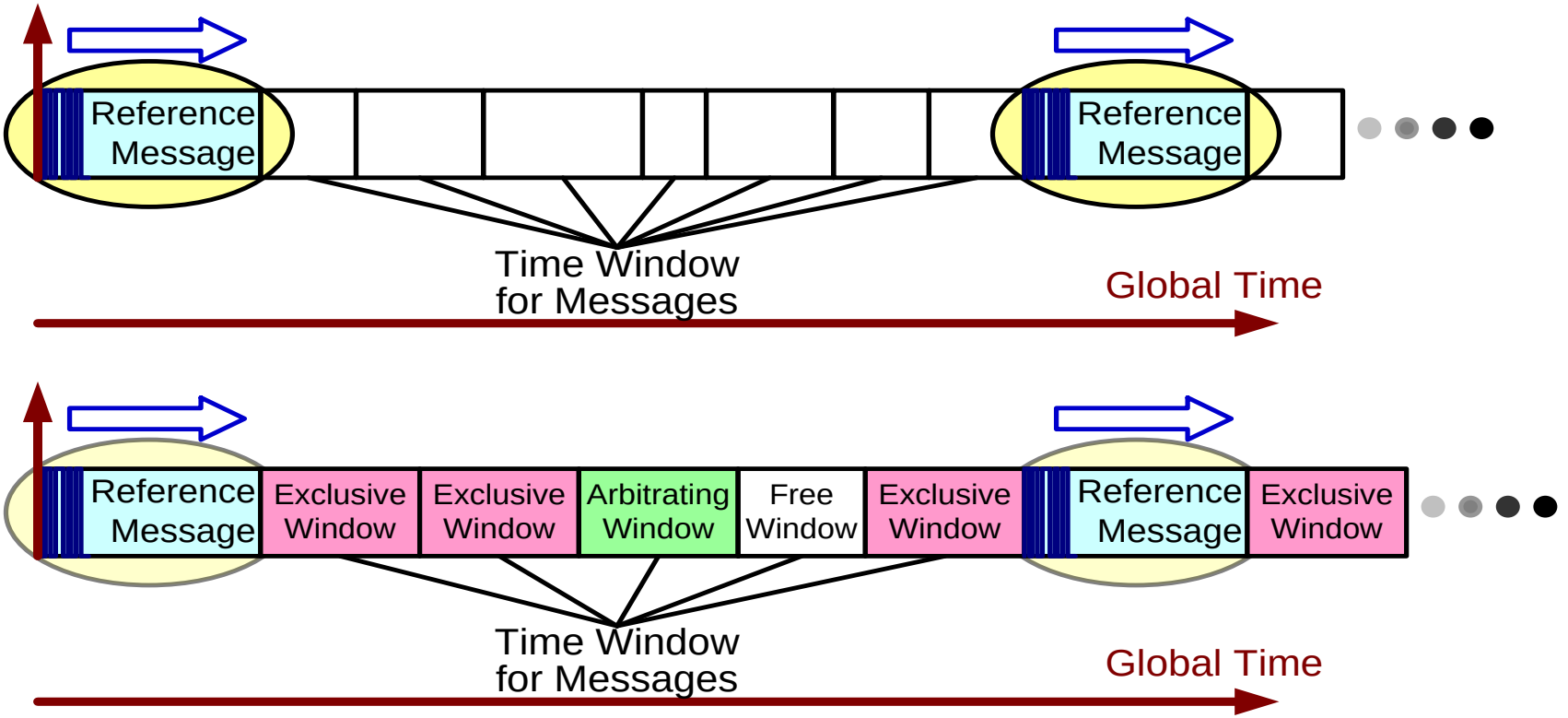
Global Time Clock Base



Global Time: CAN Towards TTCAN



TDMA: The Reference Message Triggers the Transmission; FTDMA: Arbitrating Window, Free Window



Message Control and Arbitration of TTCAN



§5.3 网络通讯/ Network and Communication

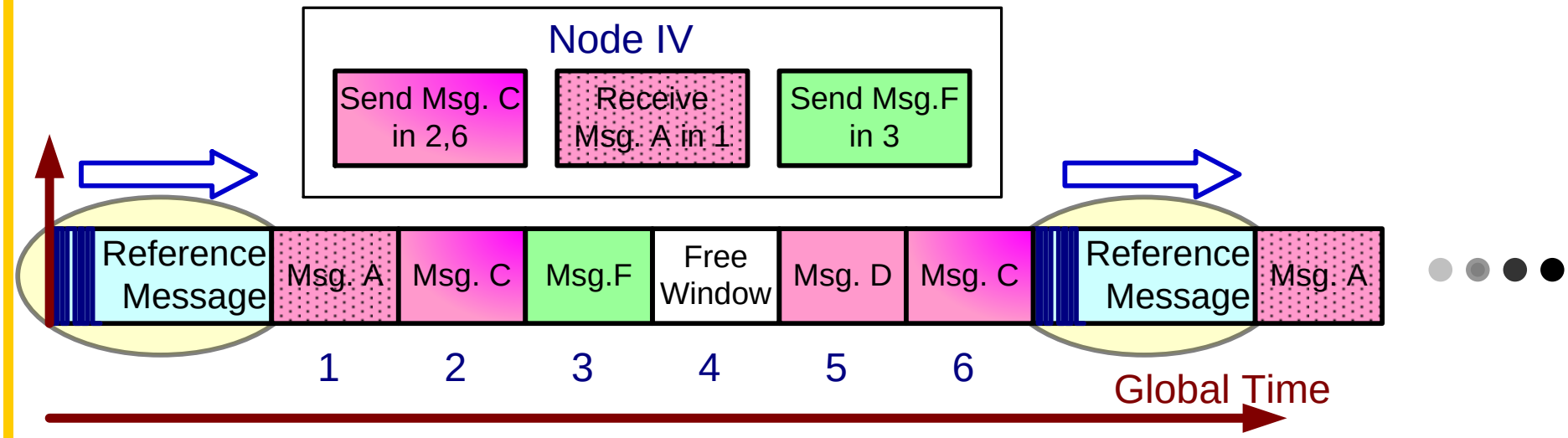
Node IV:

- ◆Send Msg. C in exclusive windows 2 and 6
- ◆Send Msg. F in arbitrate window 3
- ◆Receive Msg. A in exclusive window 1.

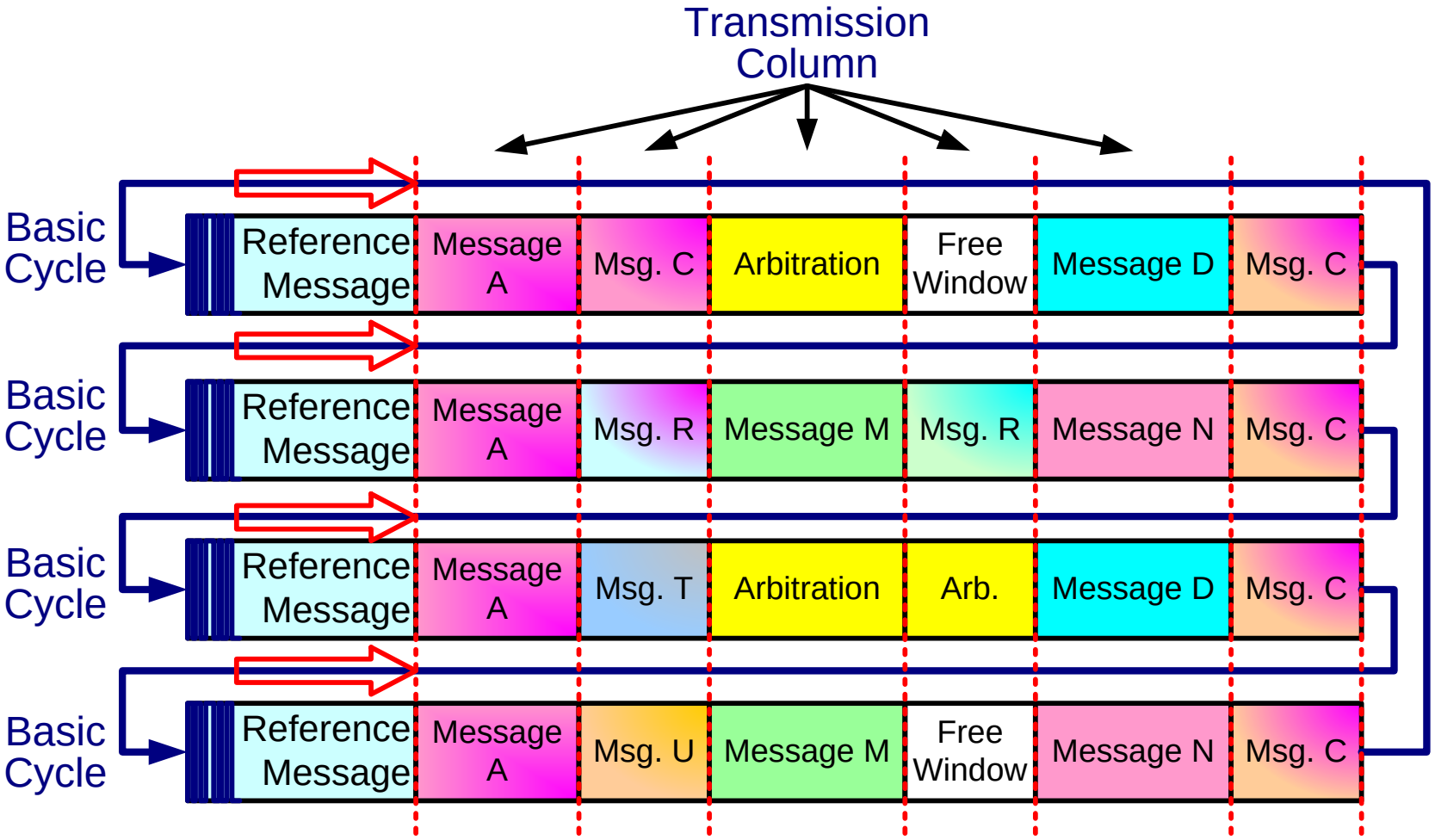
The controller only interested in receive Msg. A in time window 1. It only need add very few information to the controller.

Highly optimized memory utilization in minimizing hardware realization with enough information for Network Management.

→ High Flexible



Extension of General CAN to TTCAN

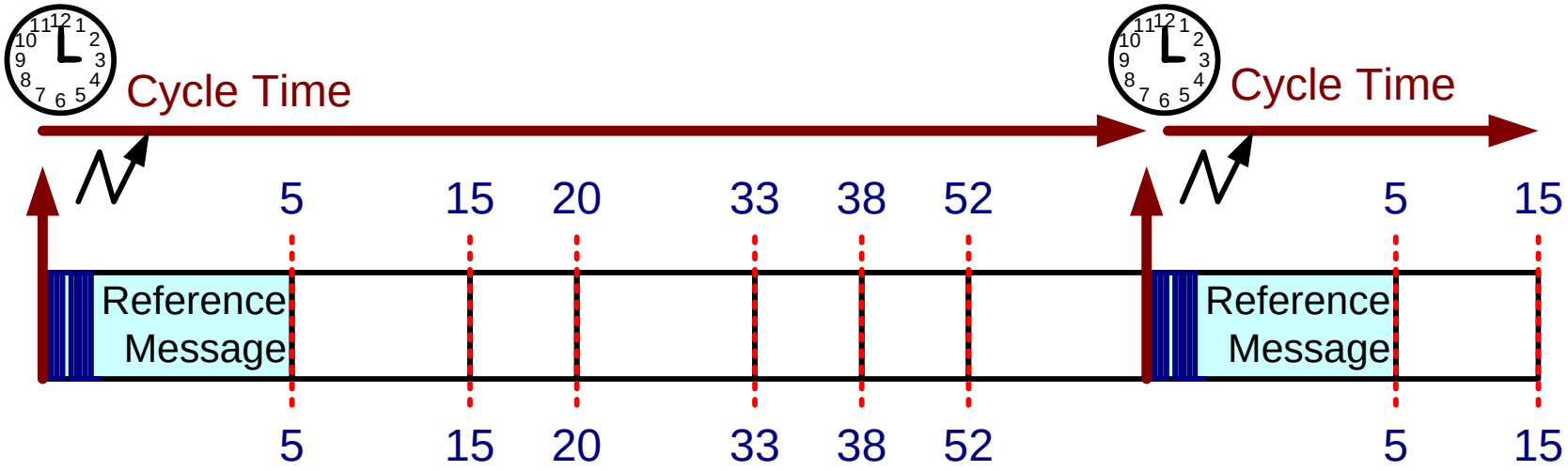


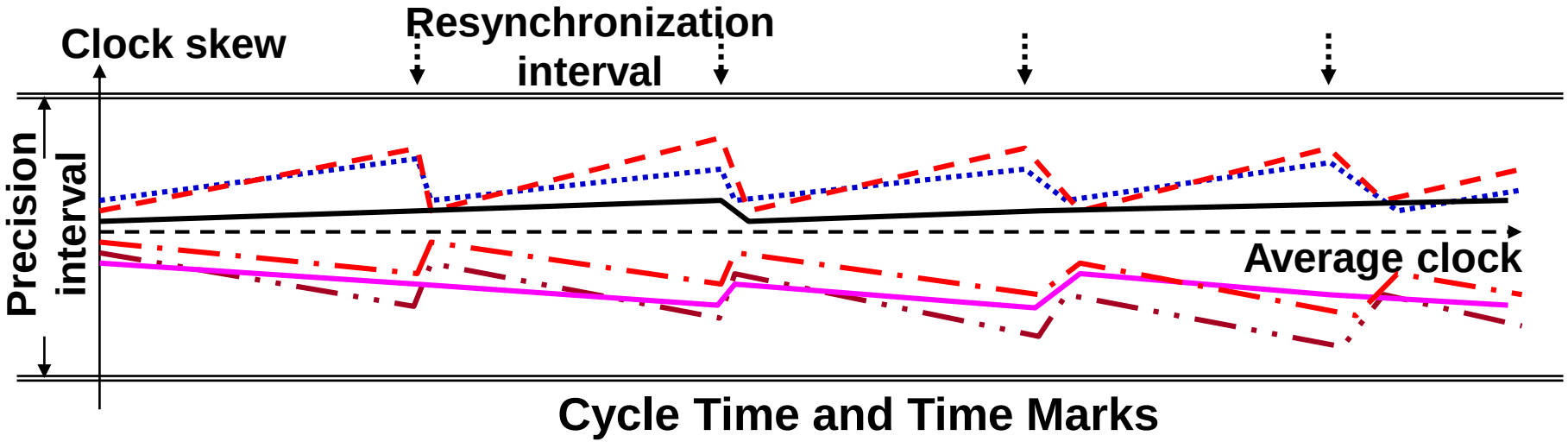
TTCAN Message Matrix Composed of Several Basic Cycles



§5.3 网络通讯/ Network and Communication

●Global Time and Time Marks





Cycle Time and Time Marks



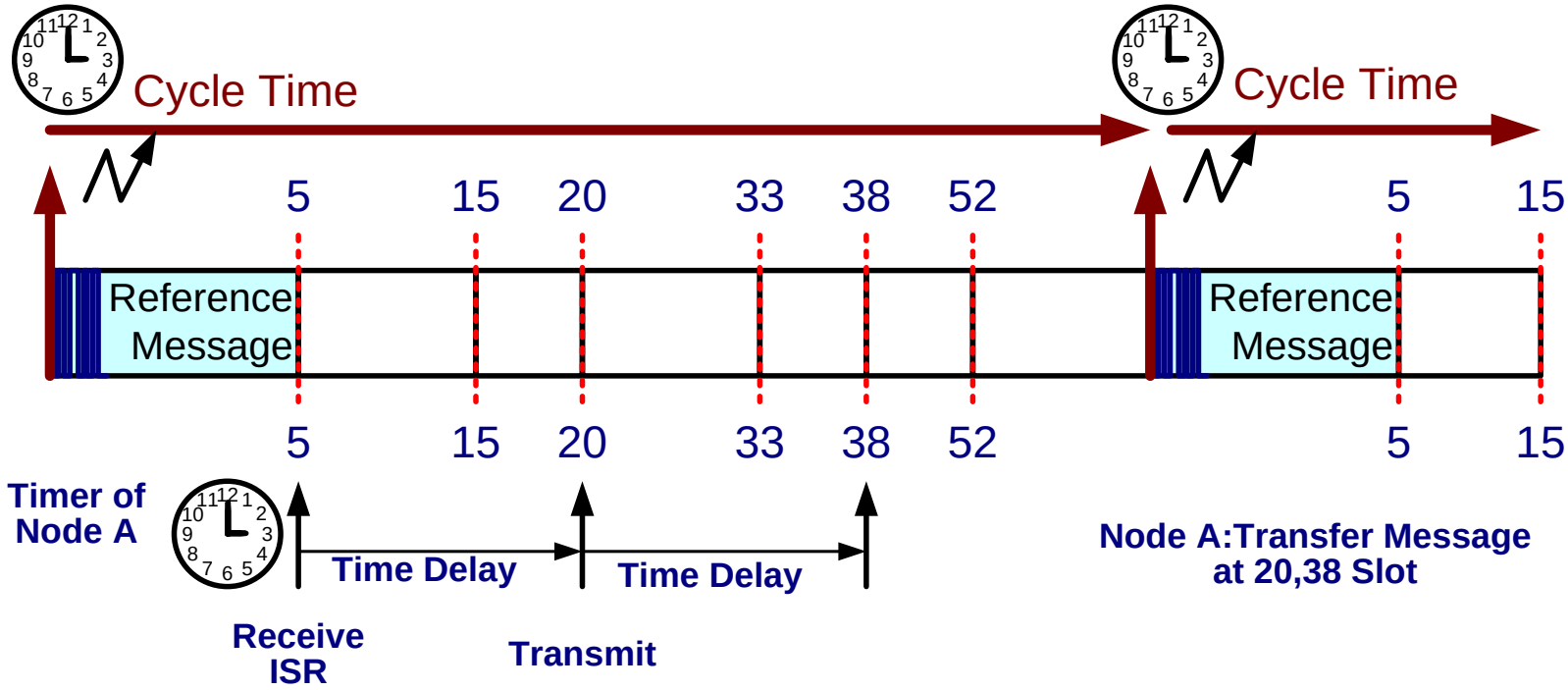
§5.3 网络通讯/ Network and Communication

Hardware Requirement:

- Same Physical Layer with CAN
- An Additional Local Timer

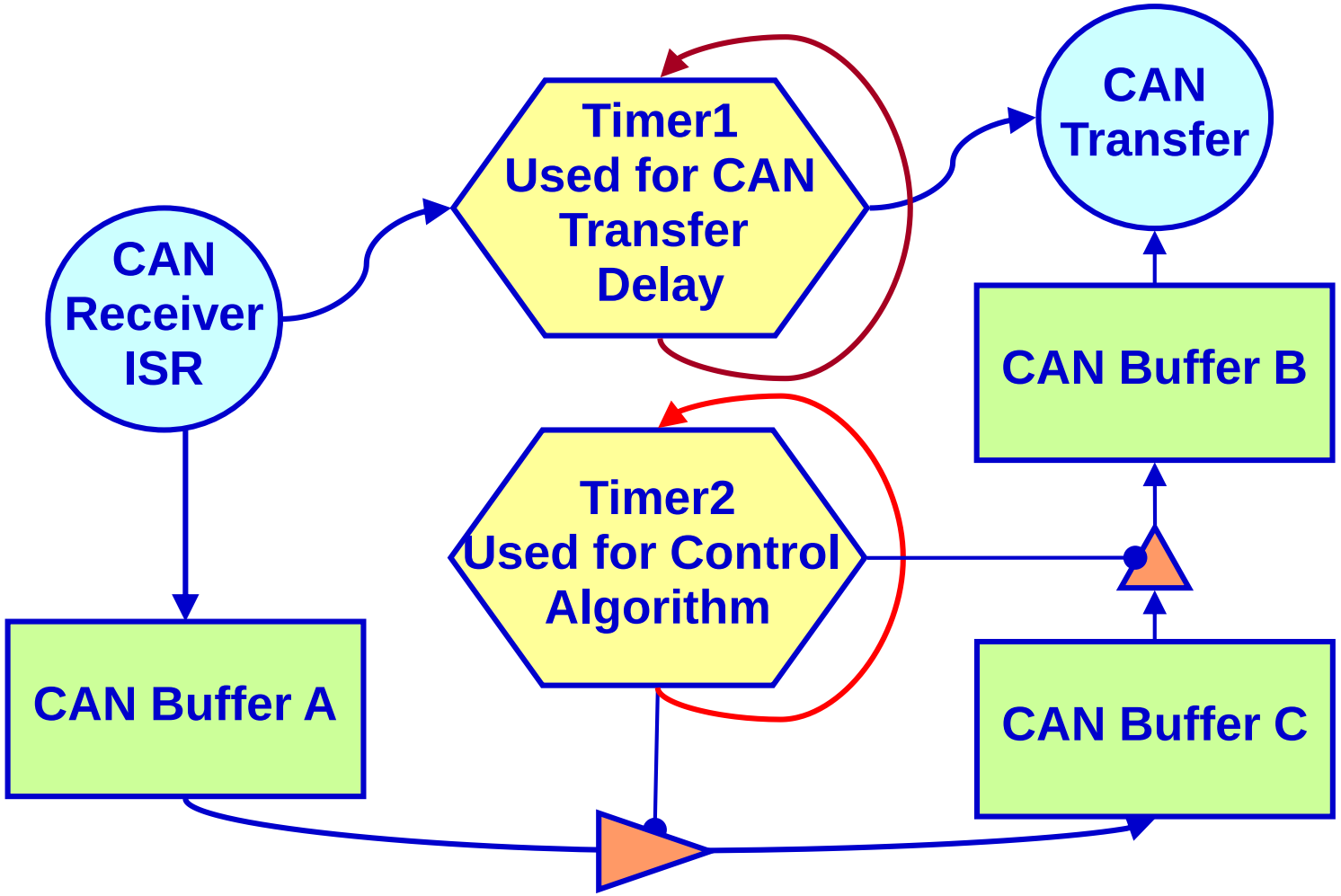
Software Requirement:

- ISR for Receiver
- ISR for Timer
- Function for Transmit
- Buffers for Message Storage





●Software Design of TTCAN

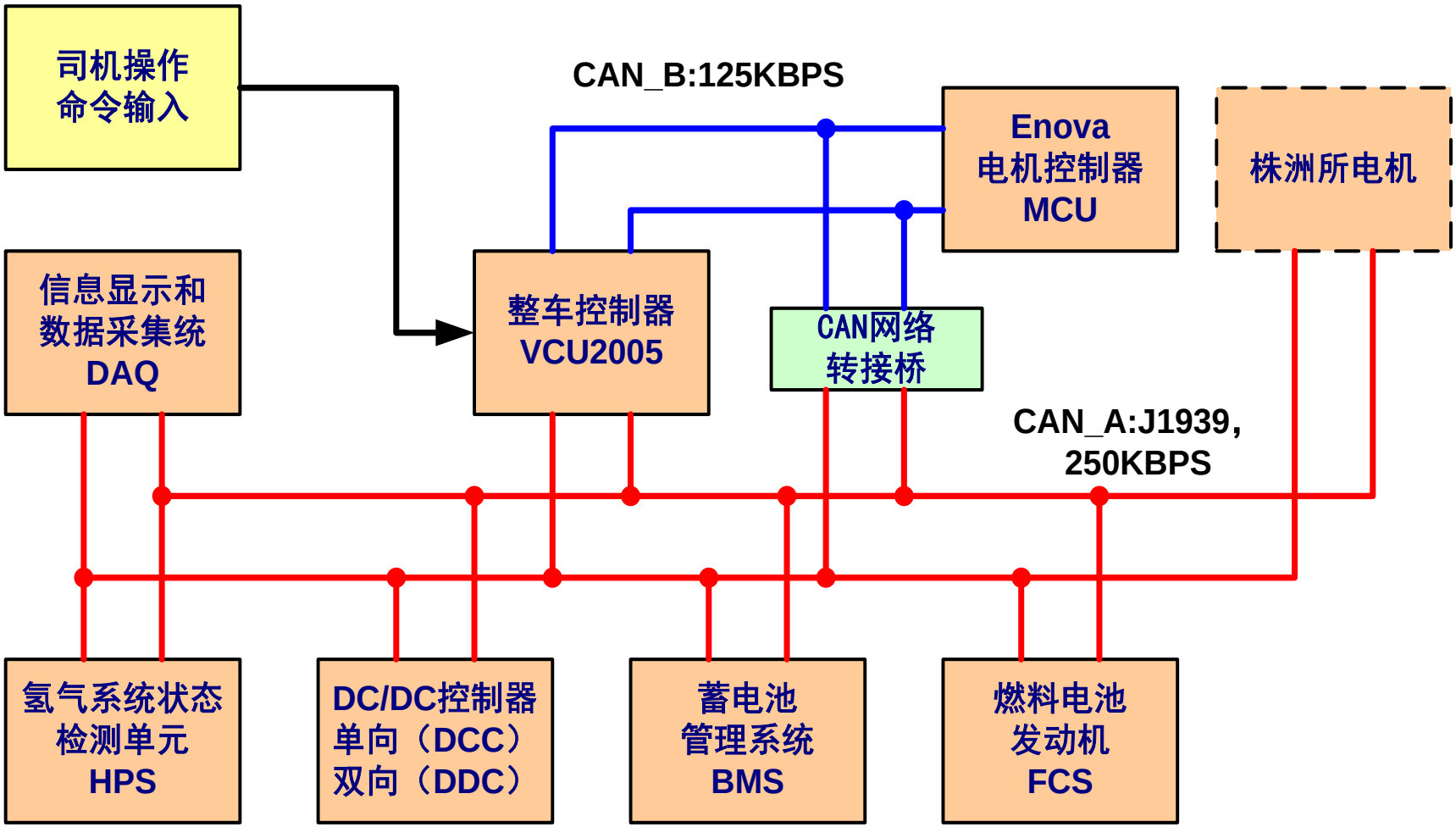




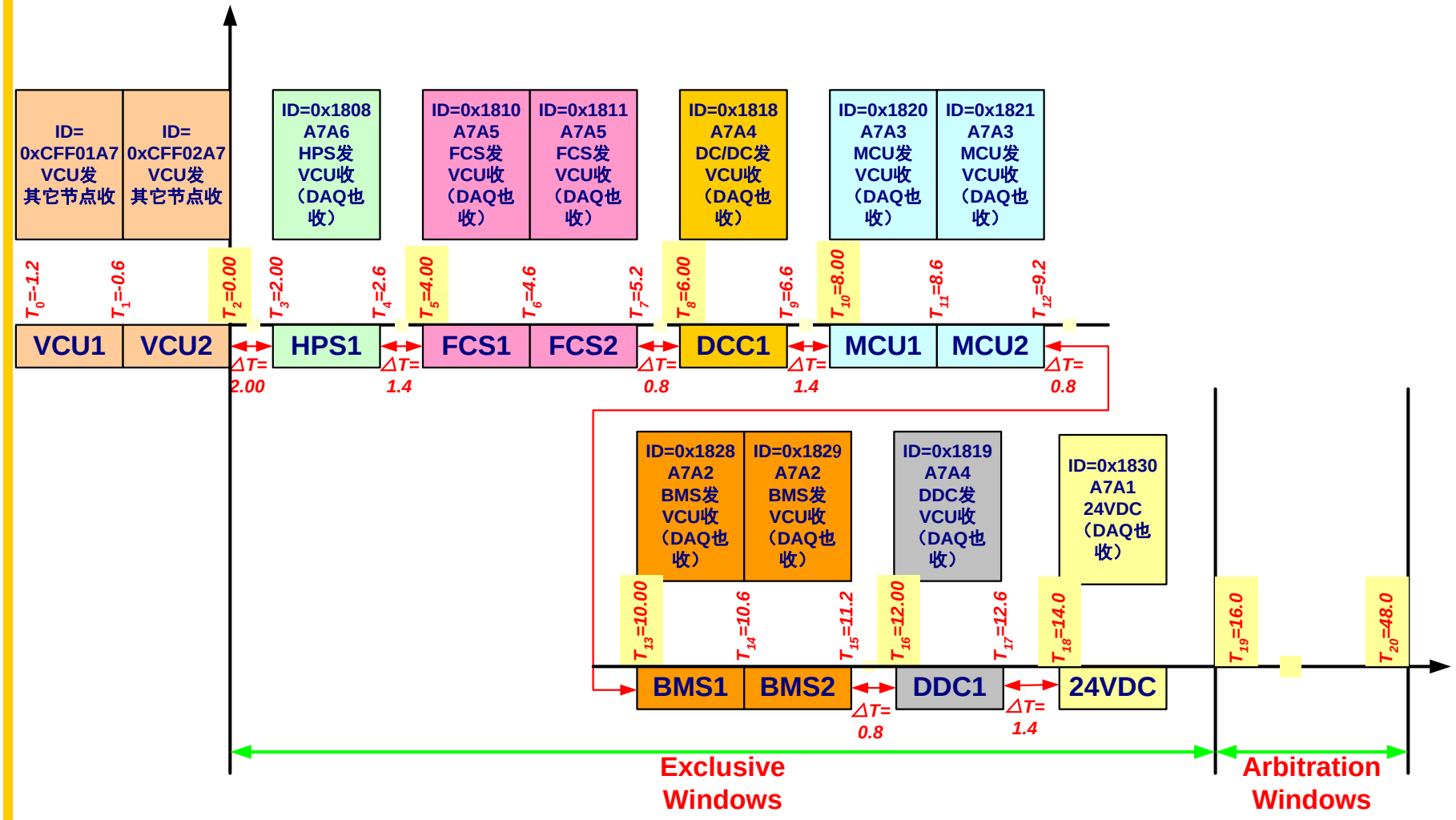
§5.3 网络通讯/ Network and Communication

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- TTCAN Protocol
- Applications of TTCAN

TTCAN in the FC Bus



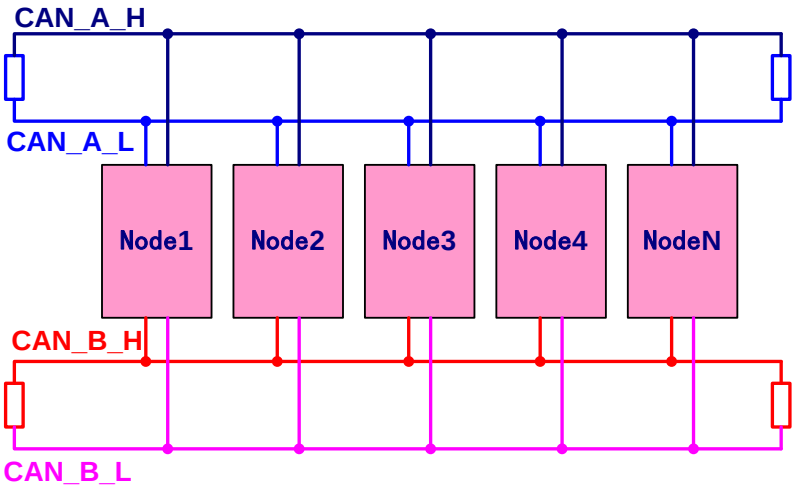
TTCAN Messages in the FC Bus

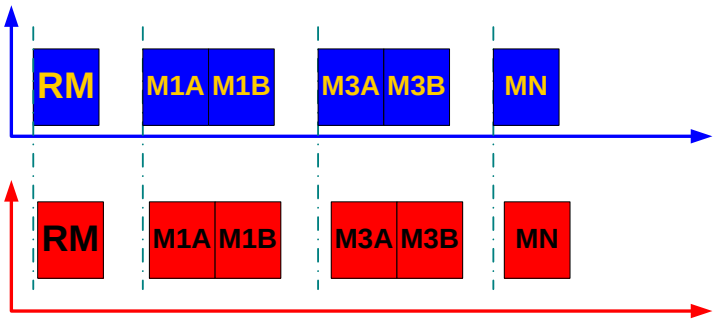


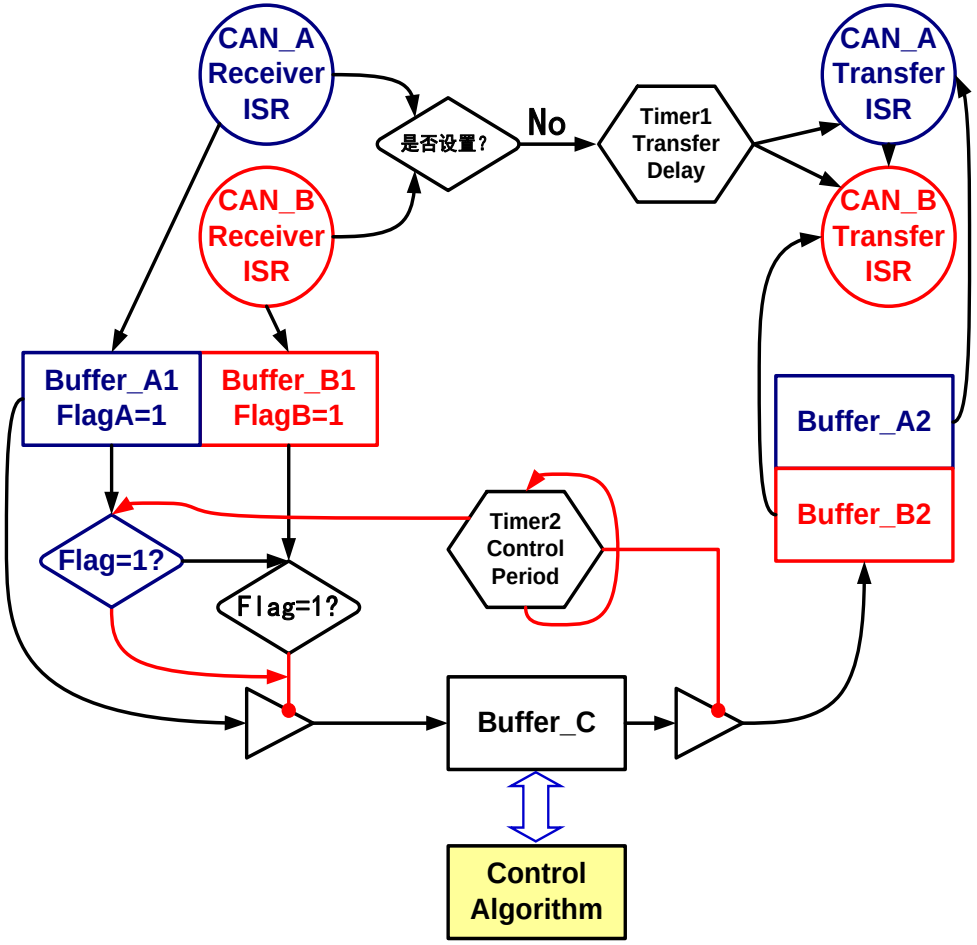


§5.3 网络通讯/ Network and Communication

Case1: Channel Redundancy









§5.3 网络通讯/ Network and Communication

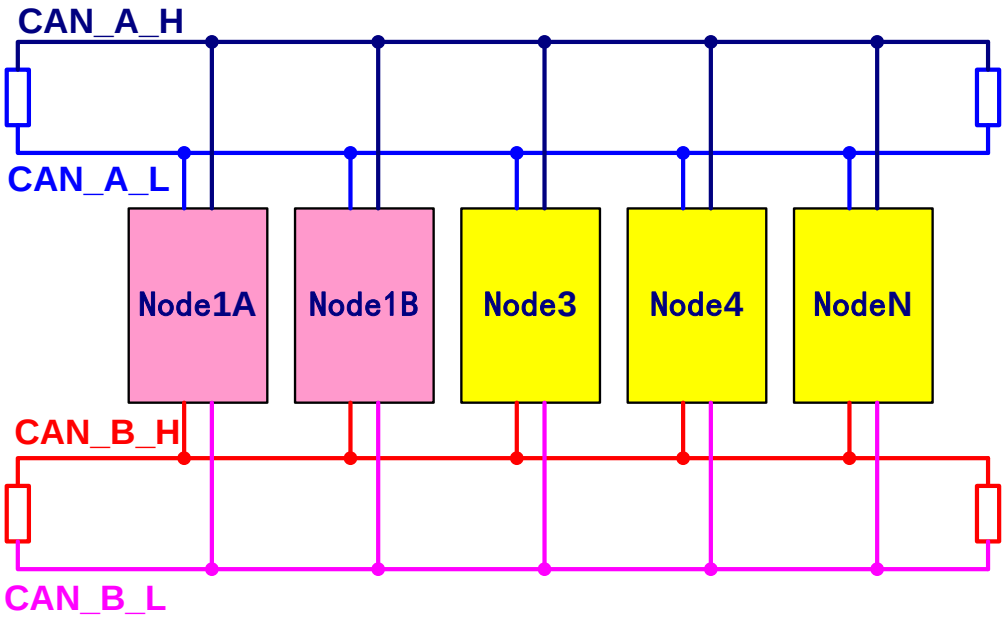
- Example: CAN_A and CAN_B are Assigned in Each Side of the Tank

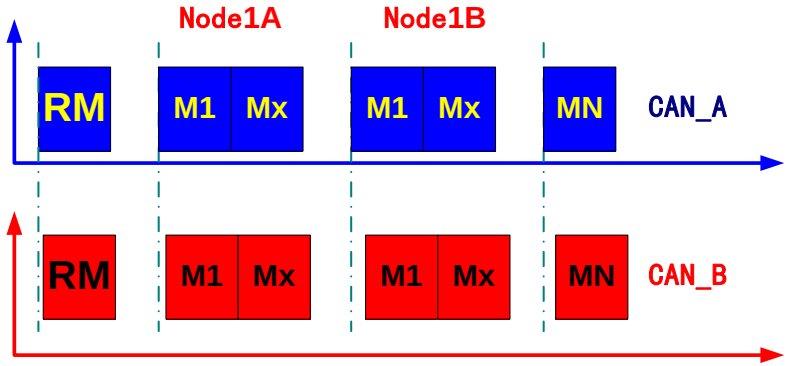


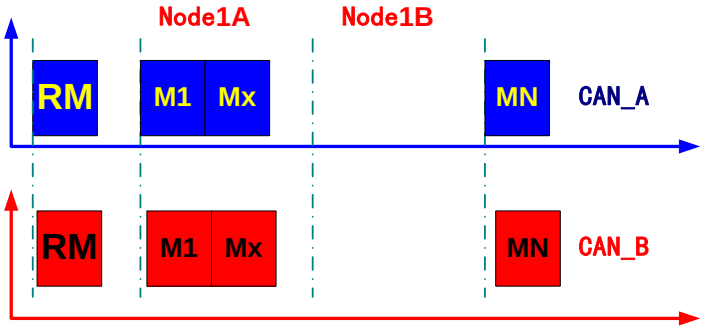


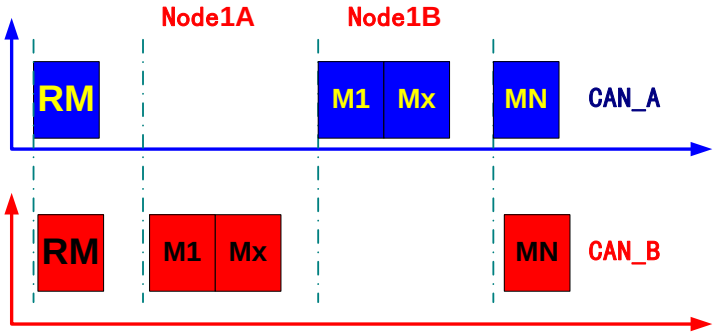
§5.3 网络通讯/ Network and Communication

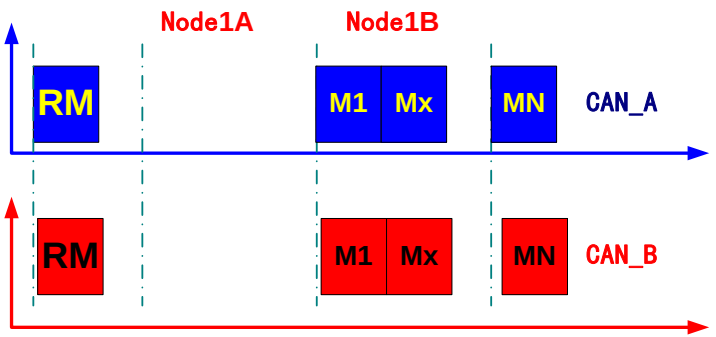
●Case 2: Bus Redundancy + Node Redundancy











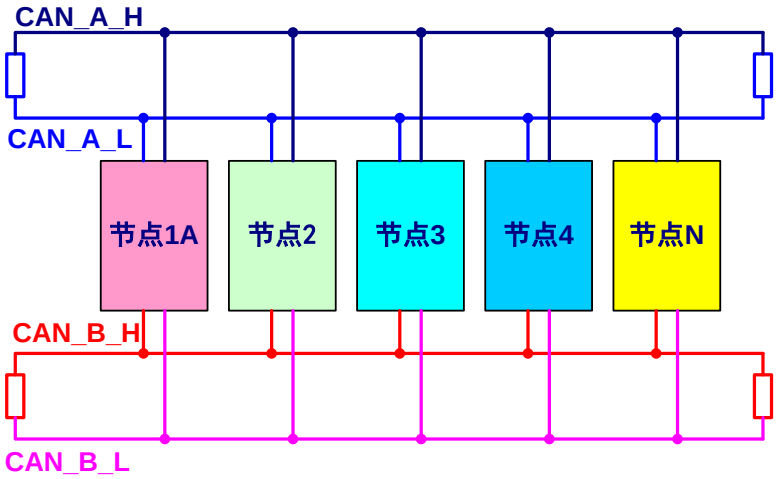


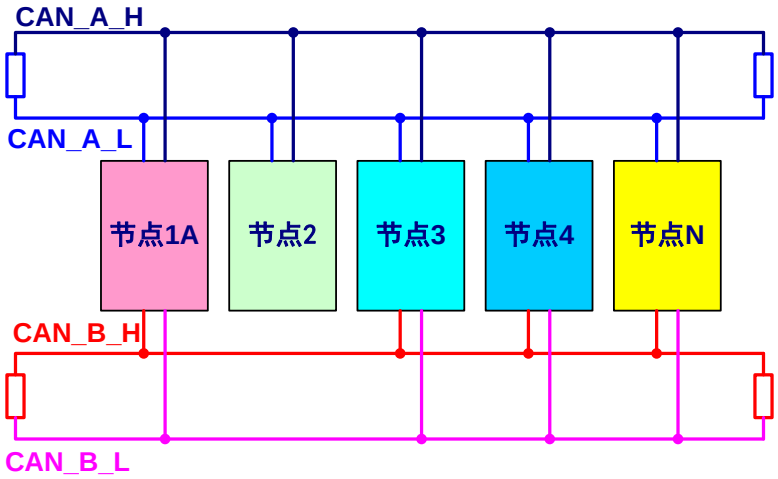
§5.3 网络通讯/ Network and Communication

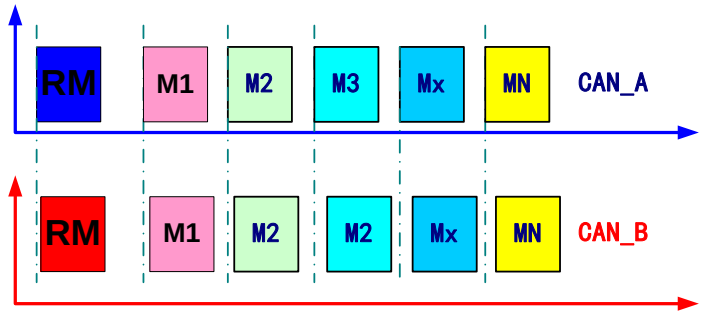
One Node Damaged: No Influence to Other Nodes!

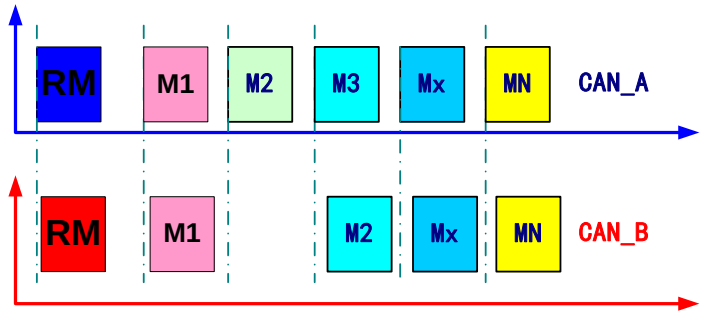
One Node Hot Plug : No Influence to Other Nodes!

→ Repair & Replacement Are Quickly During the Battle!





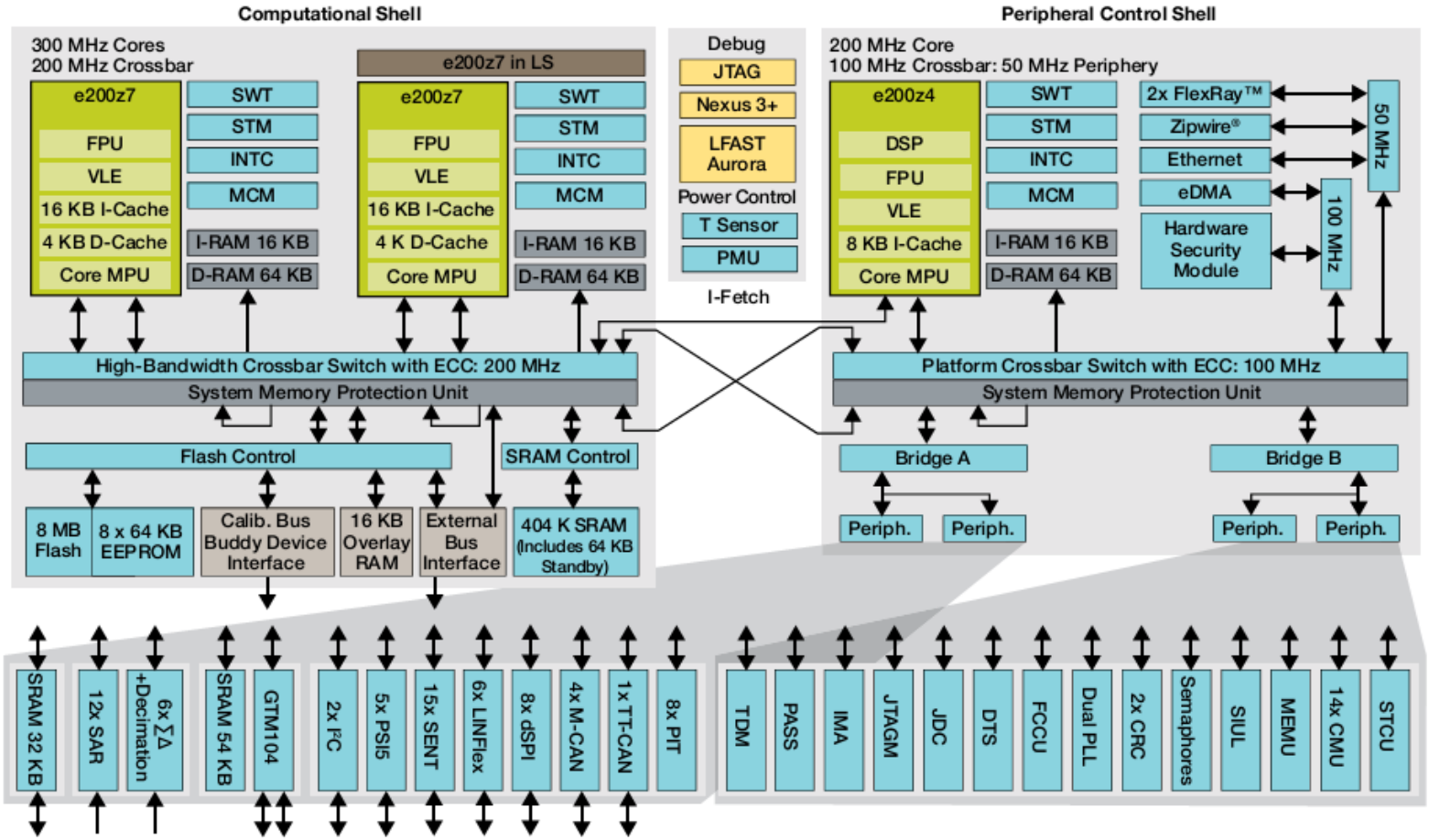






§5.3 网络通讯/ Network and Communication

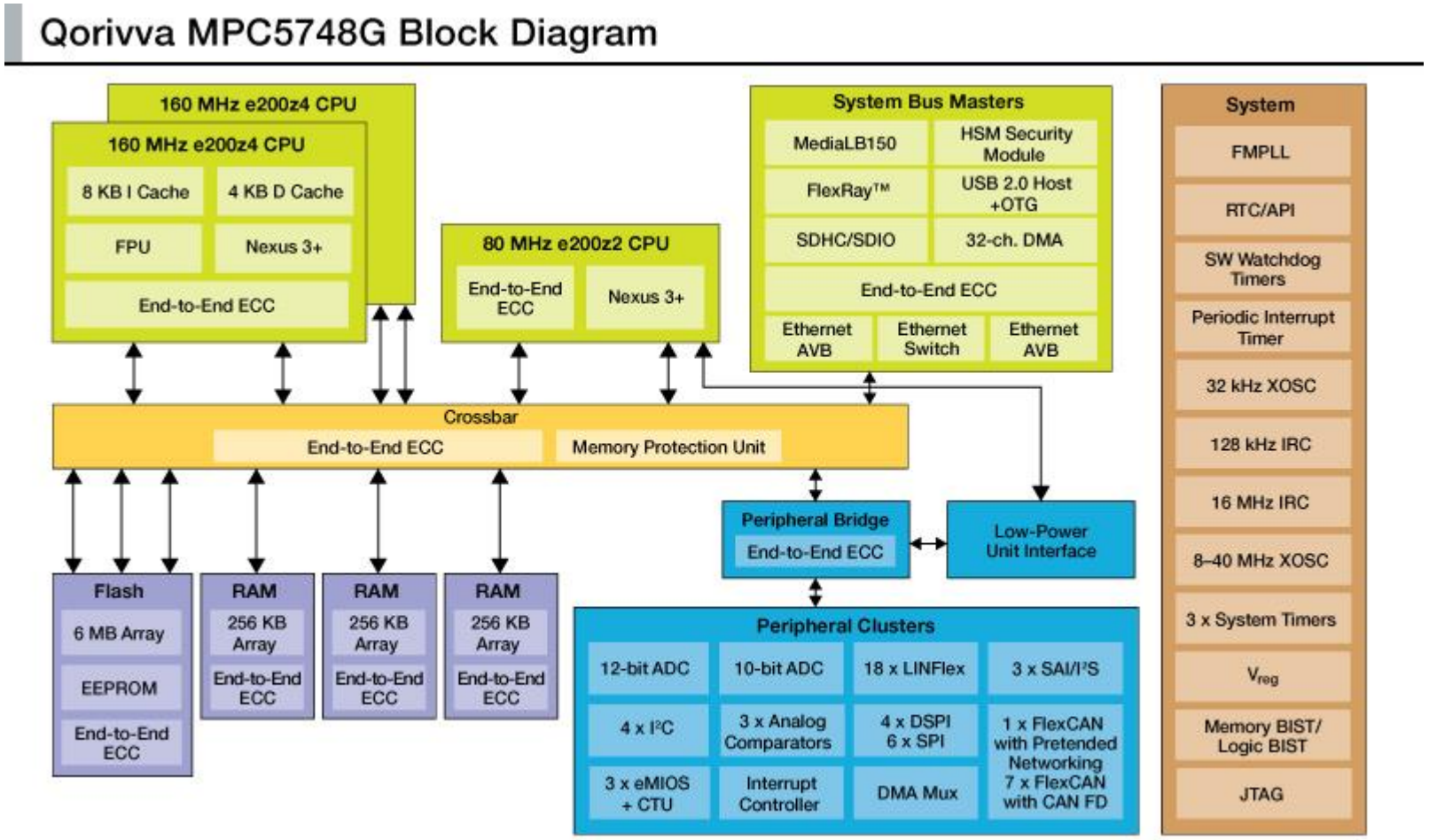
- **Multi-Core:**
  - **MPC5777M--Targeting ISO 26262 and ASIL-D, Freescale**





§5.3 网络通讯/ Network and Communication

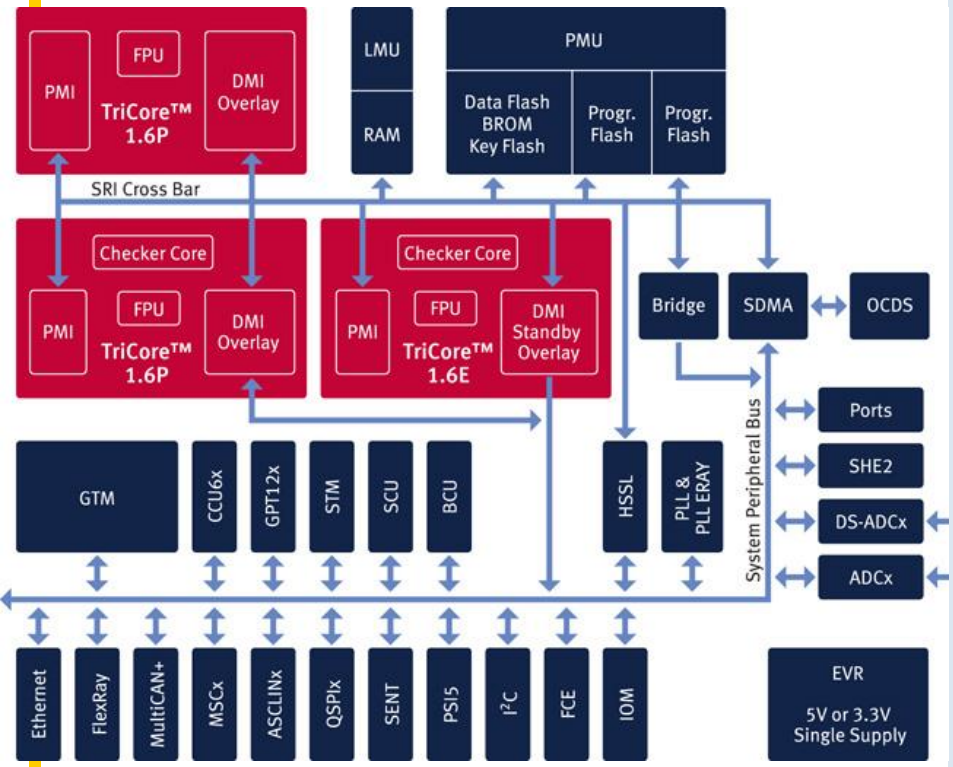
- **Multi-Core:**
  - ◆**MPC5748G—新一代网桥芯片**

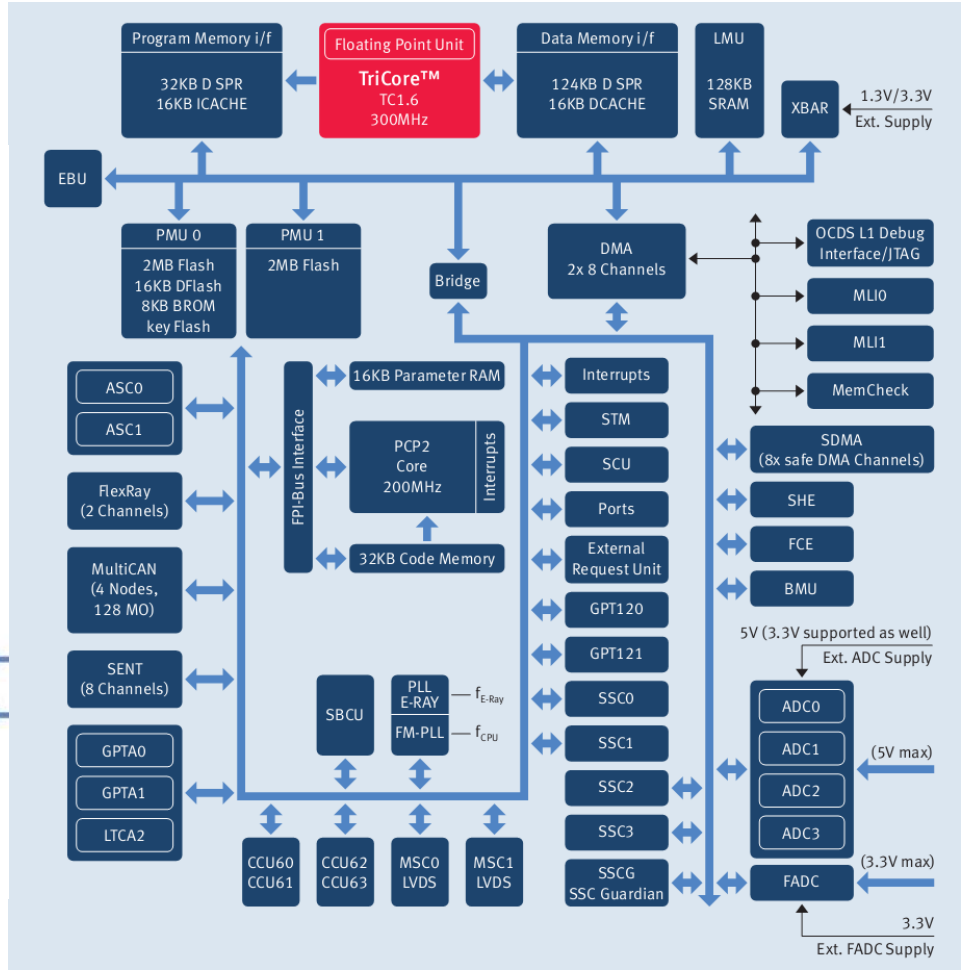




§5.3 网络通讯/ Network and Communication

- **Multi-Core:**
  - 4 Core in one Chip, Infineon.







Thank you !

(For more information, please choose my course: Mechatronic Systems In Automotive Engineering, Spring Term, 3 credits)